

Matter and Chemical Change

Picture yourself sitting around a campfire after a day of hiking. In this peaceful setting, chemical changes are taking place all around you.

- The light and heat of the campfire are evidence of a common chemical reaction.
- When you roast a marshmallow, the sugars in it react with oxygen to form a black and brittle carbon crust.
- When you eat the marshmallow, the chemical energy in the sugars is transformed into other forms of energy.
- During the day, the evergreen trees use the Sun's energy and carbon dioxide in the air to make food for themselves.

Your tent and camping gear are products of chemical reactions. You likely got to the site in a motor vehicle that used chemical reactions to move.

Chemical change is involved in every part of your daily life. In this unit you will learn more about these chemical changes, and about the choices you can make to minimize chemical changes that damage the environment.

Useful Chemistry

Getting Ready...

- How has chemistry made life easier over the past 100 years?
- What would gardening be like without fertilizers and pesticides?
- What do a bathing suit, fire extinguisher, and climbing rope have in common?
- How can you use chemicals safely?



These everyday products all have a chemical connection.

Over the past 100 years, advances in chemistry have provided many consumer products that have helped to change our lives. These advances include the ability to produce synthetic materials that have helped us reduce our need for natural materials. Instead of wool from sheep and fur from wild animals, we now use nylon, polyester, and other synthetic fibres. This has helped reduce the risk of extinction for many fur-bearing species.

Synthetic dyes have made it possible to produce dramatic colours not possible with natural dyes. For example, the orange colour used on hunting vests and clothing for emergency crews is synthetic. So are most of the colours that were used in making textiles during the twentieth century.

Chemistry has also given us many of the conveniences we take for granted. Products like detergent, Teflon™, and margarine exist because of what society has learned about chemistry.



Figure 1.1 Farmers and gardeners in the nineteenth century used manure from their livestock to fertilize the crops and gardens. Some modern farmers still use manure, but most use fertilizers produced in the chemical laboratory. Instead of relying on animal and plant waste of varying quality, chemists mix raw chemicals (such as potassium nitrate) together to make fertilizer. These manufactured fertilizers can be more effective than natural ones because they are designed for specific crops and specific field conditions. On the other hand, manure returns valuable organic matter to the soil.

Chemistry on the Farm

If you had been alive in Canada 150 years ago, you probably would have been living on a farm. As well as feeding animals, shovelling manure, and driving a team of horses, your jobs would have included pulling weeds and picking insects off the crops in your garden.

Today, only about 2.5 percent of people in Canada live on farms. Advances in chemistry and technology enable modern farmers to produce a lot more food with less human labour. Many farms have automatic feeders and barn cleaners. Teams of horses have been replaced by tractors and other machinery that allow each farmer to do the work of several people.

In addition, **synthetic** or human-made chemicals produced in a lab have reduced the amount of work farmers have to do. Instead of pulling weeds and killing insects, farmers now use synthetic **pesticides**. There are two common kinds of pesticides. **Herbicides** kill unwanted plants, and **insecticides** kill unwanted insects.

The use of chemical pesticides and synthetic fertilizers means the land can produce more food. Advances in technology and chemistry help the produce survive the trip to the consumer. In addition, the produce lasts longer on the grocery store shelves and in your cupboard.

DidYouKnow?

Pesticides such as Dursban™ are commonly used to control insects. Dursban™ is a form of organophosphate. It works by interrupting the electrochemical process that nerves use to communicate with muscles and with one another. Farmers have to handle the pesticide with great care because it can also damage the human brain and central nervous system.

Chemistry for Cleaning

One hundred years ago, farmers made their own soap using ashes from the cooking stove and natural fats from various animals living on the farm. Similar methods have been used for thousands of years.

In the past, beef fat was used when making soap. Today, manufacturers have replaced some or all of the animal fat with plant oils. For example, coconut oil may be used to make a soap that has a nice scent and cleans very well. Other ingredients are sometimes added to provide more suds or help soften skin.

Many cleaners include strong bases. Bases produce soap when they react with the fatty acids in grease. Soap consists of a two-ended molecule. One end of the molecule dissolves in grease and the other dissolves in water. This reaction enables us to wash away grease.

READING check

How does modern soap differ from the soap made 100 years ago?

Polar end:
Mixes with water

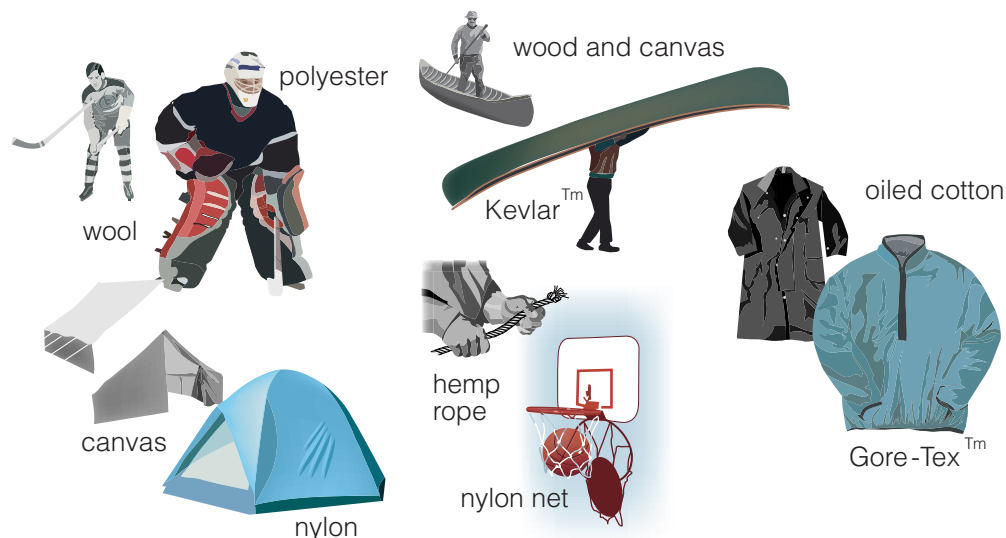
Non-polar end:
Mixes with fats



Chemistry and Synthetics

One of the major changes in the past 100 years has been the development of **synthetic fibres** or threads. One hundred years ago, clothing was made from **natural fibres** such as cotton, flax, wool, and silk.

Identify the natural and synthetic materials in these pictures.



Natural Fibres

- The downy, white fibres found in the seed pod of the cotton plant are used to make cotton thread. This is woven into cotton material.
- The stems of the flax plant are processed into threads that are woven to make linen.
- The woolly coats of animals such as sheep and goats are spun into wool. This is woven into woolen material.
- Fibres from the cocoons of silk worms are made into silk thread, which is woven into silk cloth.

Today, we still use each of the natural forms of cloth.

Synthetic Fibres

The first of these synthetic fibres was **nylon**. Nylon is made from coal, water, and air. It was first discovered by Wallace Carothers in the 1930s. When nylon was first made, scientists did not see any practical use for the material until some chemists at the company were playing games to see how far they could stretch it. They found that the stretched nylon fibres were very strong and silky.

Other, modern synthetic fibres include polyester, **Gore-Tex™**, and **Kevlar™**.

Did You Know?

Polyester is a fibre that bounces back into shape quickly and does not absorb very much moisture. This material is used to make clothes that dry quickly and do not need ironing.

Check Your Understanding

1. How have chemicals made modern farming easier than it was 100 years ago?
2. How has the process of making soap changed over the past 100 years?
3. Name two synthetic and two natural fibres. Describe at least two advantages of synthetic fibres.

Key Terms

synthetic
pesticides
herbicides
insecticides
synthetic fibres

1.2 Plastics, Polymers, and Alloys

Another major change over the past 100 years has been the development of **synthetic polymers**. **Polymers are a type of compound that occurs in nature.** For example, the silk in spider webs and the sap from rubber trees are two natural polymers.

During the past 75 years, chemists have made synthetic polymers in the laboratory. The great thing about synthetic polymers is that scientists can develop different ones with physical characteristics that meet a particular need. You know many of these synthetic polymers as types of plastic.

Table 1.1 shows some synthetic polymers, their properties, and how they are used. Read the descriptions and identify ways that you use each of them.

Table 1.1 Synthetic Polymers and Their Uses

Polymer	Significant Properties	Uses
neoprene	rubbery, porous, insulating, does not react readily with other chemicals	shoe soles, hoses, wet suits
polyethylene	can be easily formed into lightweight containers	milk jugs
polystyrene	does not absorb water, poor conductor	styrofoam cups, insulation
polyurethane	flexible, fibrous, does not absorb water	upholstery, clothing

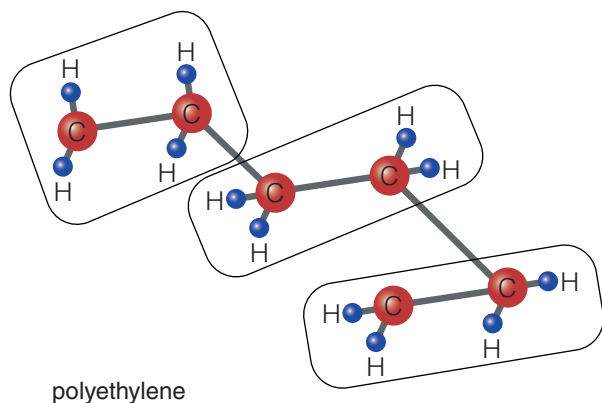


Figure 1.3 A polymer is a giant molecule made up of many small and identical sub-molecules. A polymer is put together like a string of beads. In the case of polyethylene, each bead consists of two carbon atoms and four hydrogen atoms. In this visual, the circled areas show one unit of the polyethylene polymer.



Synthetic polymers can help you see. The lightweight lenses in today's glasses are made from polycarbonates. These lenses are much lighter and can be made much thinner than old-fashioned glass lenses.

Find Out **ACTIVITY**

Make Your Own Polymer

In this activity, you will learn how to make a synthetic polymer using substances you can find at home.

Safety Precautions



- Clean up the work area and wash your hands thoroughly at the end of the activity.

What You Need

disposable cup
stirring rod
graduated cylinder
5 mL white glue
5 mL water
10 mL borax solution

What to Do

1. Mix 5 mL of water with 5 mL of white glue in a disposable cup.

2. Add 10 mL of borax solution to the glue mixture and stir for 2 min.

3. Roll the lump in your hands until it is not sticky anymore.

What Did You Find Out?

1. (a) Reshape the putty into at least three forms.
(b) What happens when you try to bounce or stretch the putty?
(c) What else can you do with it?
2. How have the physical characteristics of the substances changed throughout the activity?
3. This product was around for a long time before anyone thought of using it as anything except a toy. Suggest two or three other uses for it.

SKILLCHECK

Initiating and Planning

☀ Performing and Recording

Analyzing and Interpreting

☀ Communication and Teamwork

READING check

Explain what an alloy is.
Give one example.

Alloys

Have you ever wondered why stainless steel does not rust? In the past, cooking utensils were made of iron that rusted. Modern kitchen utensils are made from several metals mixed to look like one metal. **This type of mixture is an alloy.** Alloys have properties of the original metals that make up the alloy.



Figure 1.4 Many kitchen utensils are made of stainless steel. Stainless steel is a mixture of iron, chromium, nickel, and carbon.

Using Alloys

Many of the products you use today — from cars to jewellery — are made from alloys. Steel for buildings is an alloy of iron and chromium. Cars made from steel in the early 1900s were very strong, but they were heavy and prone to rusting. The steel in today's cars is also mixed with chromium. This steel alloy has the strength and durability of iron, plus chromium's resistance to rusting.

A new type of steel has been developed called ultrahigh-carbon steel. This new steel contains more carbon than was used in previous steels. Ultrahigh-carbon steel can be stretched to 100 times its length without breaking. Earlier forms of steel could be stretched to only 1.5 times their original length.

This new steel can be used to make structures that are light but still very strong.



Figure 1.5 Silver and copper in these gold rings provide strength. The gold jewellery you wear is actually a gold alloy — a mixture of gold and another metal, such as silver, copper, zinc, or nickel.

Pure gold is very soft and gets damaged easily, so most jewellery is made with alloys. For example, 14k gold is actually an alloy of gold, silver, and copper.

DidYouKnow?

Brass is an alloy made from copper and zinc. Musical instruments such as trumpets, horns, and cymbals are usually made from brass.

Check Your Understanding

1. What metals are found in stainless steel utensils? How would your life be different if kitchen utensils were not made from stainless steel?
2. In the winter, you choose clothes with fabrics that will keep you warm.
 - (a) What materials keep you warm today?
 - (b) What materials would you have used 100 years ago?
3. Synthetic polymers are used in many products. Name three synthetic polymers and briefly explain how they have changed daily life.

Key Terms

synthetic polymers
alloy



Figure 1.6 Many hazardous substances are found at home. Some of these substances are stored in a cupboard to keep them away from young children. Others are kept in special containers that prevent accidental poisoning.

Almost all of these products have safety warnings on their labels. Hazardous Household Product Symbols (HHPS) are designed to be easy to understand. The shapes that outline the pictures are designed to look like road signs.

READING check

Explain how one cleaner uses chemical change to clean. What is the negative side of this chemical change?

Chemicals in the Cupboard

If you did a quick survey of the cupboards in your kitchen, bathroom, and laundry room, you would probably find all-purpose cleaners, toilet-bowl cleaners, and a variety of other cleaning products. Cleaners usually contain substances that are reactive. They start chemical changes that break down the dirt and make cleaning less strenuous.

Drain cleaners, for example, often contain highly corrosive substances such as sodium hydroxide. Sodium hydroxide is a base used to make drain and oven cleaners. Bases like sodium hydroxide help clean clogged drains by breaking down the proteins (such as skin cells and hair) and animal fats that cause the problem. Heat is given off, which helps melt the grease.

Table 1.2 Common Chemicals and How They Clean

Chemical	Where Found	How It Cleans
muriatic acid (hydrochloric acid)	• steam-iron cleaner	reacts with lime (scaly buildup) to form a soluble product
acetone	• nail polish remover	dissolves nail polish so it can be wiped away
methanol (wood alcohol)	• paint solvent	dissolves the paint

Try ↓ This!

Many of today's furniture polishes include hazardous substances. In the nineteenth century, people polished their furniture using products that were less hazardous. You can make your own furniture polish by mixing 20 mL of lemon juice with 1 L of mineral oil. Spray or wipe this mixture onto wooden furniture and wipe it with a dry cloth.

The mineral oil will seep into the wood and help prevent dust from sticking to the wood. The lemon juice will help speed up the absorption.

Chemistry and Cooking

A knowledge of chemistry is important to chefs, bakers, and cooks. Knowing how different ingredients react allows them to control chemical reactions. For example, bakers may use yeast to make bread. **Yeast is a single-celled organism that needs food, warmth, and moisture to thrive. It converts its food (sugar and starch) into carbon dioxide and alcohol. Carbon dioxide bubbles make baked goods rise.**

A mixture of vinegar and baking soda will also produce carbon dioxide gas. Sometimes bakers use this mixture to make baked goods rise. Sponge toffee, for example, is a mixture of vinegar, baking soda, and sugar.



Figure 1.8 Yeast is responsible for the chemical reaction that makes bread rise. This knowledge helps bakers produce different kinds of bread.

DidYouKnow?

Cooking involves many other chemical reactions. For example, citrus fruits react with kitchen utensils made of aluminum. The acid in these foods reacts with the aluminum in the utensils and produces hydrogen gas. This reaction will cause the utensil to become corroded and no longer useful.

To avoid such a reaction, many chefs use kitchen utensils made from stainless steel. Stainless steel will not react with acid.

The inside of an aluminum soft-drink can does not react with the acid in the soft drink because the aluminum has been coated with a plastic polymer that keeps the pop from coming into direct contact with the aluminum.



Figure 1.9

Check Your Understanding

Key Terms

HHPS

1. Many household cleaners rely on chemical changes to do their work. What chemical change occurs when you add drain cleaner to a clogged pipe?
2. Identify three chemicals commonly found in household cleaners and describe how each of these chemicals helps to clean.
3. (a) Yeast helps bread to rise. Explain the chemical change that yeast produces and how this makes bread rise.
(b) If you wanted a flat bread, would you use more or less yeast? Explain.

1.4 Chemistry at Work

Even 100 years ago, chefs were using chemistry to do their work. So were blacksmiths and welders. For example, the welder in Figure 1.10 knows that pieces of steel must be heated to a very high temperature to melt them. Only then can they be welded together. A chemical change involving acetylene gas and oxygen produces enough heat to melt steel.

In the early 1900s, blacksmiths served as both mechanics and welders. When they were not shaping metal, they often experimented with different ways to combine metal and other materials.

Today chemistry is used in every workplace. The list below provides some examples.



Figure 1.10 The acetylene torch uses a chemical reaction that produces carbon dioxide, water, and enough energy to melt steel.

Automotive Repair Shop — Paints, fillers, and cleaners all involve chemical reactions. Special car paint protects metal car bodies from rusting. Waxes and polishes protect that paint. When the paint gets faded by the Sun, special waxes may remove tiny amounts of paint to restore the shiny appearance.

Hair Salon — Just about everything but the scissors involves a chemical change. This includes colorants, curling solutions, shampoos, and conditioners.

Office — Hazardous chemicals are common in inks, toner cartridges, correction fluid, and the fire retardants in paper, upholstery, and carpets.

Construction Site — Hazardous chemicals are found in demolition explosives, wood preservatives, paints for finishing, and solvents for cleaning up. Many solvents used to clean grease stains or wash windows include ammonia. Ammonia works by dissolving grease, wax, and window dirt.

READING Check

Every workplace uses some type of chemical reaction. Name one chemical reaction used where you work part-time or in your classroom.

Cosmetic Chemistry

Figure 1.11 Do you use shampoo that adds texture to your hair? If so, you are putting plastic in your hair. Such shampoos simply cover hair with plastic resins. The plastic resins make the hair thicker, which appears to improve its texture.



READING check

Soap will clean your hair, but why won't soap make your hair shiny and smooth?

Before 1938, most people cleaned their hair with soap. Have you ever used bar soap on your hair? It probably left your hair dull and rough — not really the look you wanted.

This happens because soap is a base that breaks down protein such as hair. Soap also reacts with calcium in water, leaving a scum layer on your hair. This layer makes hair dull and difficult to comb.

Even before 1938, hair stylists knew how to deal with this layer of scum. They used an acid rinse made from vinegar or lemon juice to dissolve the scum, making the hair shine again.

When you buy shampoo, you are buying a synthetic detergent that will help wash oil and dirt from your hair. Dish detergents wash grease and dirt from your dishes.

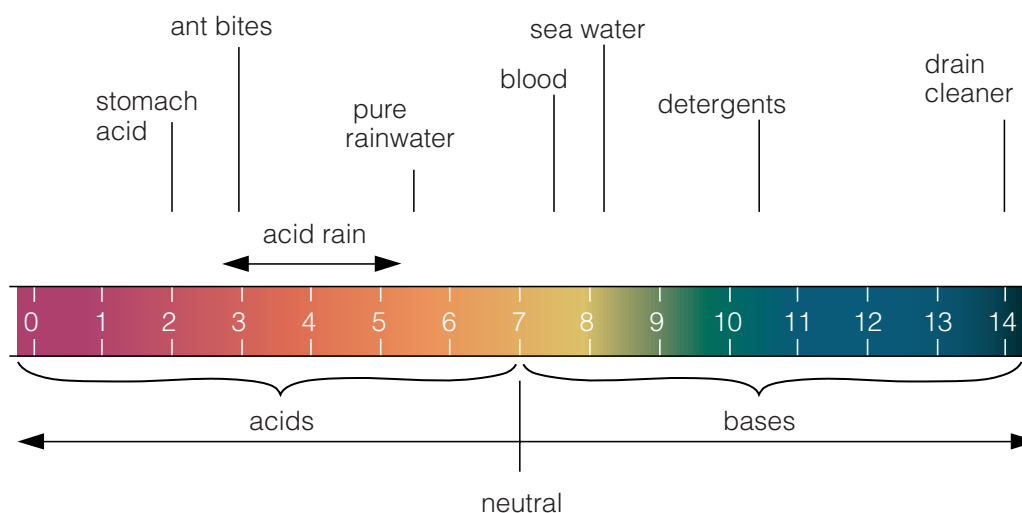


Figure 1.12 This diagram shows the pH of some common substances.

READING

check

How is chemistry useful in the development of cosmetics? Explain.

Make-up and Face Cream

Cosmetics have been around for thousands of years. Some of these cosmetics are harmless, but others are not.

Many cosmetics are tested on animals before being used on people. Researchers study how much the animals are affected before allowing the material to be used on humans.

Today, many groups oppose the use of animals in testing cosmetics. As a result, many manufacturers advertise that their products have not been tested on animals. Knowledge of chemistry helps manufacturers control chemical reactions. Use the next activity to make your own skin cream.

Find Out ACTIVITY

Make a Skin Cream



Cosmetic manufacturing is big business today. Each year, consumers spend billions of dollars on everything from creams and oils to shampoos and shaving cream. Many of these products use chemical changes to keep you looking good.

In this activity, you will make a hand cream using readily available materials. As the skin ages, the outer layers become thinner and dryer. This tends to exaggerate fine lines and wrinkles.

Various skin creams will help make the skin look smoother by putting moisture back into the skin and creating a barrier to prevent evaporation. Of course, the best way to make your skin look good is to eat a proper diet and drink plenty of water.

Many skin creams are emulsions of oil, water, and wax. An emulsion is a mixture of two liquids, such as oil and water, that normally do not mix. If you shake oil and water together, you get an emulsion that is unstable and will separate quickly.

An emulsifier can be used to make the emulsion more stable. An emulsifier works by coating the droplets of one liquid so they stay suspended in the other liquid.

Safety Precautions



- Clean up the work area and wash your hands thoroughly at the end of the activity.

What You Need

60 mL distilled water
120 mL mineral oil
1 g borax
30 mL grated beeswax
hot plate and wire mesh screen
ice bath or access to a refrigerator
100 mL beaker
250 mL beaker
1000 mL beaker or small saucepan
100 mL graduated cylinder
stir stick or plastic spoon

What to Do



1. Use the graduated cylinder to add 60 mL of distilled water to the 100 mL beaker.
2. Dissolve the borax in the distilled water and set aside.
3. Use the graduated cylinder to add 120 mL of mineral oil to the 250 mL beaker. Add the beeswax to the mineral oil.

4. Pour 100 mL of tap water into the 1000 mL beaker. Place this on the hot plate and gradually warm the water. This is the hot water bath.
5. Carefully place the 250 mL beaker of oil and wax mixture into the hot water bath. Heat the mixture for 10 minutes or until the beeswax melts.
6. Put on heat-proof gloves and carefully remove the melted oil-wax mixture from the hot water bath. Remove the hot water bath from the hot plate. Let the hot plate cool down before doing the next step.
7. Place the 100 mL beaker containing the borax solution on the hot plate and bring it almost to a boil.
8. Slowly add the borax solution to the oil-wax mixture. Stir this mixture continuously.
9. Cool the beaker containing the oil, wax, water, and borax mixture in an ice bath or the refrigerator.
10. Check the beaker every 5 minutes for the remainder of the class. Note the appearance of the mixture.

The Key to Chemical Safety

The acid from a car battery will burn your skin. That is why people wear gloves when handling a car battery.

How do you know what safety precautions to use when handling a substance you have never used before? The best way is to read the label.

If the dangerous substance is found in a workplace — including a school, restaurant, or repair shop — you will find a **WHMIS** label on the container. **WHMIS** stands for **Workplace Hazardous Materials Information System**. It provides information on how to store, handle, and dispose of dangerous substances at work.



Figure 1.13 Look for these WHMIS symbols on any product you use at work.

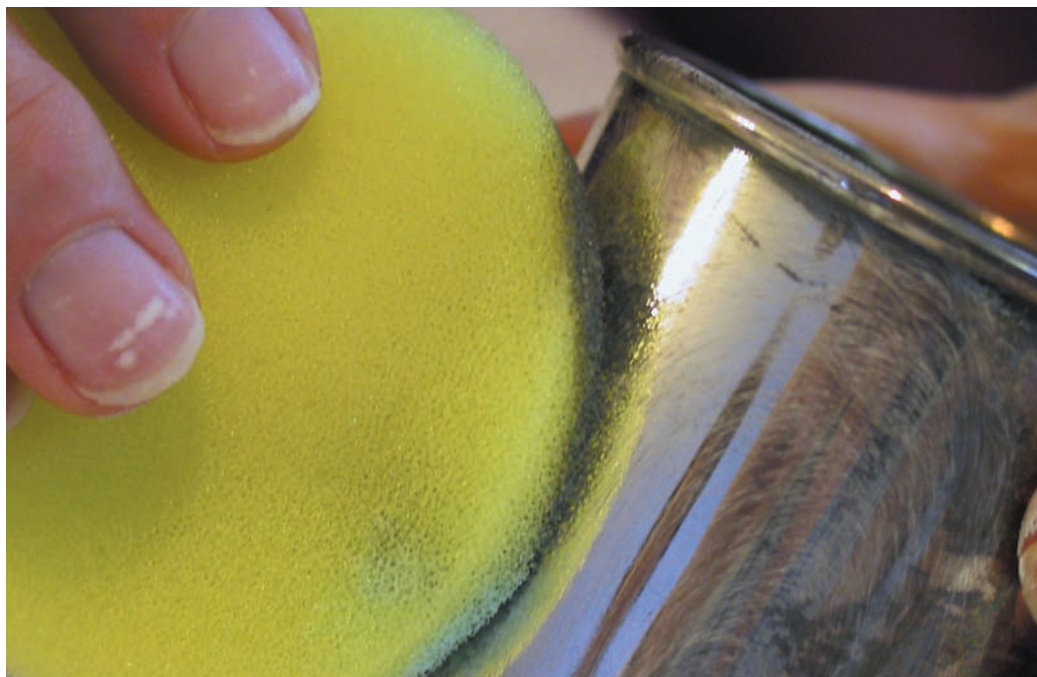
If you are working with hazardous materials, you should also check the **MSDS** or **Material Safety Data Sheet**. The MSDS provides detailed information that will not fit on the product label. **Following is a sample MSDS.**

MATERIAL SAFETY DATA SHEET - 9 SECTIONS	
SECTION 1 - PRODUCT INFORMATION	
Product Name Product Use Manufacturer's Name Physical and Mailing Address Emergency Contact Phone Number	WHMIS Classification (optional) Supplier's Name Physical and Mailing Address Emergency Contact Phone Number
SECTION 2 - HAZARDOUS INGREDIENTS	
Hazardous Ingredients (very specific)	
SECTION 3 - PHYSICAL DATA	
Physical State (What does it look like? Is it a liquid, gas, or solid?) What happens to it under a variety of circumstances? (i.e. heat, freezing, dropping, etc.) Flammability and how to extinguish. Includes a wide variety of details concerning how easily this product	
SECTION 4 - FIRE AND EXPLOSION DATA	
will ignite / explode and how to deal with it. How stable is this product?	How it reacts under various conditions.
SECTION 5 - REACTIVITY DATA	
Incompatibility with other substances. Information about how the product affects and enters the body. Immediate affect. Long term toxic affect.	Hazardous Decomposition Products
SECTION 6 - TOXICOLOGICAL PROPERTIES	
Exposure limits. In summery, immediate and long term affects to the human body.	
SECTION 7 - PREVENTIVE MEASURES	
Personal Protective Gear; ventilation, etc.; leak and spill info; waste disposal; handling and storage; special shipping instructions	
SECTION 8 - FIRST AID MEASURES	
Information for immediate first aid treatment. Usually always ends with "contact a Doctor"	
SECTION 9 - PREPARATION INFORMATION / Who prepared this and contact info	

Common Reactions

Getting Ready...

- How do chemical reactions make your life easier?
- Why must you take special care when dealing with certain chemicals?
- How is energy involved in chemical reactions?



Chemistry happens all around you. Millions of chemical reactions occur all the time, even though many of them are invisible to you.

If you could see what happens to the molecules in the flour, sugar, and other ingredients when you bake a cake, you would observe some substances breaking down and others forming. When you shampoo, condition, and colour your hair, you are taking advantage of chemical reactions.

Have you ever used silver polish to clean a favourite silver ring, necklace, or cup? All those swirls and pretty decorations make that job a challenging one — unless you know a little chemistry!

In this chapter you will take a closer look at some everyday chemical reactions and begin to appreciate their usefulness to you.



What You Will Learn

In this chapter you will learn:

- how chemical reactions are useful to you
- how substances change during a chemical reaction
- how to describe different types of chemical reactions
- how to identify the evidence of a chemical reaction
- that energy is absorbed or released by all chemical reactions

Why It Is Important

- You breathe chemicals, eat chemicals, sleep on something made of chemicals, and use chemicals to make your life easier. Knowledge of chemical reactions can help you understand how substances can help or harm you. This knowledge can make you a wiser consumer and provide you with a more comfortable lifestyle.

Skills You Will Use

In this chapter you will:

- collect and organize information so it is easy to interpret
- demonstrate a knowledge of WHMIS standards by applying proper handling techniques
- identify new questions or problems from what you have learned

2.1 Useful Reactions



Figure 2.1 The space shuttle has separate tanks of liquid hydrogen and liquid oxygen. When ignited, these produce a chemical reaction that fuels the shuttle.



Figure 2.2 Ripening bananas go through a chemical reaction. What tells you that a reaction has occurred?

Almost everything you do involves a **chemical reaction**. During a chemical reaction, substances react to produce new substances with different properties from the original substances. Some of these reactions are simple, and some are complex. For example, launching a space shuttle uses complicated technology. A simple chemical reaction makes it possible.

Hydrogen and oxygen combine during a chemical reaction to produce energy and water. Energy powers the space shuttle, and water is a by-product of the reaction.

On the other hand, an everyday occurrence like taking an antacid to treat acid indigestion is based on a different type of chemical reaction. The antacid tablet contains a basic chemical called magnesium hydroxide. Magnesium hydroxide reacts with the hydrochloric acid in your stomach to produce neutral magnesium chloride and water.

How about an even more common chemical change? When you leave a banana on the kitchen counter for a few days, it turns from green to yellow to brown. It is obvious that changes are occurring, as shown in the flow chart that follows.

bananas release natural ripening gas known as ethylene → ethylene triggers enzyme production → enzymes speed up chemical reactions

- 1 → enzyme amylase breaks down starches into sugar
- 2 → other enzymes trigger colour change and softening of fruit
- 3 → enzymes break down large molecules into smaller ones that evaporate in the air, providing the banana aroma



What element gives bleach its whitening power?

Whitening Whites

Household bleach helps us keep white laundry looking bright and clean. The active component in some bleaches is hydrogen peroxide. Other bleaches contain sodium hypochlorite. Both of these key ingredients contain oxygen. Oxygen is responsible for the reaction that makes a coloured stain colourless.

Baking Bread

In Chapter 1, you learned about the role of yeast in making bread. Baking bread is a lot like working in a chemistry lab. A baker combines flour, yeast, liquid, salt, and other flavourings in a bowl. The ingredients are shaped into a dough that is left to rise, and then the dough is baked in a hot oven to produce a tasty, nutritious loaf of bread.

Flour is made of cereal grains that release starch and proteins when they are milled. **When flour is mixed with water and yeast, enzymes in the yeast break down the starch into glucose. Other enzymes react with some of the glucose molecules to produce carbon dioxide and ethanol, an alcohol. The carbon dioxide then bubbles up through the mixture, causing the dough to rise.** Heat causes the pockets of gas in the dough to expand, giving the bread its airy texture. The alcohol burns off in the baking process, leaving behind a delicious flavour.



Figure 2.3 A lot of chemistry goes into every loaf of freshly baked bread.

Swimming Pool Scent



Figure 2.4 Chemical reactions involving chlorine keep pools clean and disinfected for swimmers.

You have probably noticed that swimming pools smell distinctly of chlorine.

When chlorine is added to water, it produces hypochlorous acid.

Hypochlorous acid kills micro-organisms by destroying their cell membranes and inner enzymes.

Cleaners that include chlorine always have warning symbols. That is because chlorine gas is dangerous when inhaled, and it can lead to unconsciousness or even death. Protective rubber gloves and eye protection should be worn when working with chlorine.

DidYouKnow?

The digestion of food requires chemical reactions. **Digestive enzymes break down the proteins, carbohydrates, and fats from the foods we eat and convert them to energy for our cells to use. Digestion starts in the mouth, where the enzyme *amylase* in the saliva breaks down starches to simple sugars. In the stomach, the enzyme *protease* speeds up the breakdown of proteins into amino acids. Partially digested foods are then carried to the intestines where an enzyme called *lipase* breaks down fats into glycerol and fatty acids.**

READING Check

What substance is produced when chlorine is added to water?

Identifying Common Gases

Many chemical reactions produce oxygen, hydrogen, and carbon dioxide gases. Knowing how to test for these gases can explain what happened in a particular chemical reaction. You will not actually carry out this experiment, follow along with the procedure and attempt to predict the outcomes.

Problem

How do you determine the presence of hydrogen, oxygen, or carbon dioxide gases?

Prediction

Part 1: What will happen to the limewater if carbon dioxide gas is produced?

Part 2: What will happen to the burning splint if hydrogen gas is produced?

Part 3: What will happen to the glowing splint if oxygen gas is produced?

Safety Precautions



- When testing for gases, point the mouth of the test tube away from yourself and other people.
- Clean up the work area and wash your hands thoroughly at the end of the investigation.

Apparatus

3 small test tubes
petri dish
medicine dropper
funnel

Materials

distilled water
vinegar
5 g baking soda
50 mL limewater
20 mL hydrogen peroxide solution
10 grains dry yeast
manganese dioxide
2 wooden splints
matches

Procedure

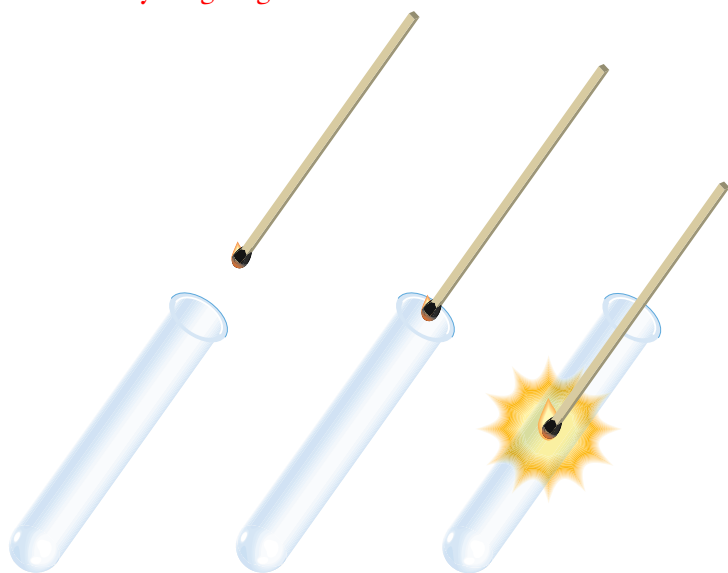
Part 1: Carbon Dioxide Gas

- 1 Fill a small test tube half full with distilled water. Add some vinegar until the liquid reaches just below the top of the test tube.
- 2 Fill the bottom of a petri dish half full with distilled water.
- 3 Add 5 g of baking soda to the test tube. Immediately turn the test tube upside down and place it into the petri dish.
- 4 Observe what happens inside the test tube. Use your thumb to stopper the test tube and then turn the test tube right side up again.
- 5 Using a medicine dropper, add a few drops of limewater to the test tube. **Limewater turns cloudy in the presence of carbon dioxide gas.**

Part 2: Hydrogen Gas

- 1 Fill a clean test tube half full with distilled water. Add hydrogen peroxide solution until the liquid reaches just below the top of the test tube.
- 2 Fill the bottom of a clean petri dish half full with distilled water.
- 3 Add about 10 grains of dry yeast to the test tube. Immediately turn the test tube upside down and place it into the petri dish.
- 4 Lift up the test tube and stopper it with your thumb.
- 5 Light a wooden splint.

- Insert the burning wooden splint into the test tube as shown. **If the gas suddenly causes a squeal or pop, this indicates the presence of hydrogen gas.**



Test for oxygen gas

Part 3: Oxygen Gas

- 1 Fill a clean test tube half full with distilled water. Add hydrogen peroxide solution until the liquid reaches just below the top of the test tube.
- 2 Fill the bottom of a clean petri dish half full with distilled water.
- 3 Add 1 g of manganese dioxide to the test tube. Immediately turn it upside down and place it into the petri dish.
- Lift up the test tube and stopper it with your thumb.
- 5 Light a wooden splint and blow out the flame so the splint is only glowing.
- Insert the glowing wooden splint into the test tube. **If the flame re-ignites, this indicates the presence of oxygen gas.**

Analyze

1. Summarize the three tests you used to test for the presence of carbon dioxide, hydrogen, and oxygen gases.
2. How did the results of the three investigations compare to the predictions?

Conclude and Apply

3. Problems may occur in the collection of gases from chemical reactions. What are some?

Extend Your Skills

4. A common product of chemical reactions is water. Suggest a simple test that could be used to test for the presence of water as a result of a chemical reaction.

Check Your Understanding

1. Describe three chemical reactions that you read about in this section.
2. List two chemical reactions you did not read about in this section.
3. Liquid bleach is sometimes used as a disinfectant in kitchens. Predict what gas might be formed as the sodium hypochlorite breaks down.

2.2 The Changes That Occur



Figure 2.6 Carefully placed explosives can be used to demolish buildings. This photograph shows the demolition of the Bow Valley Centre in October, 1998.



Write definitions for the terms “reactant” and “product.”

Chemical reactions occur all around you. Some reactions, like those involved in cooking, require energy. Others release energy.

In a chemical reaction, the substances you start with are not the same as the substances you end up with. Substances with new properties are formed.

Reactants	Products
The substances that are there before a chemical reaction.	The substances that form afterwards.

There are several indicators of chemical change. One sign is the release or absorption of energy.

In some chemical reactions:

- gases may be given off;
- new solids may be formed; or
- colour changes may occur.

Sometimes odours are also given off.

Combustion and Neutralization

Consider what happens when you light a candle. You bring a lit match or a lighter close to the candle and ignite the wick. You then observe the wick burning, and feel heat and see light being given off.

Candle wax burns in the presence of oxygen to produce carbon dioxide and water. This type of chemical reaction, in which a fuel burns in oxygen, is referred to as a **combustion** reaction. Much of Alberta's industry depends on combustion reactions involving fossil fuels such as oil and natural gas.

Another group of chemical reactions are **neutralization** reactions. During these reactions, an acid is added to a base to produce a compound and water. The pH of the product is more neutral (closer to 7) than the pH of either reactant.

READING
check ✓

List the reactants in combustion reactions and in neutralization reactions.

A very common neutralization reaction—one that occurs when you take an antacid. You can describe this chemical reaction in one sentence.

Magnesium hydroxide reacts with hydrochloric acid to produce magnesium chloride and water.

You can further simplify this using symbols. The result is known as a **word equation**.

magnesium hydroxide + hydrochloric acid → magnesium chloride + water

Some other examples of chemical reactions.

DidYouKnow?

Although toothpaste ingredients vary by brand, most contain some of the following common ingredients:

- a base such as aluminum hydroxide ($\text{Al}(\text{OH})_{3(s)}$) to neutralize the acids in your mouth that can cause cavities
- sodium bicarbonate (baking soda) to help neutralize acids
- fluoride, which bonds to the enamel of your teeth to further protect them from the acids in your mouth
- abrasives, such as calcium carbonate, to help remove stains and plaque
- detergents to create the foaming action that keeps the toothpaste in your mouth, preventing it from running out as you brush
- flavourings to cover the taste of detergent
- dyes or colourings to give toothpaste a more pleasing look. For example, titanium dioxide is often used to make some toothpastes look bright white.



An Invisible Gas Made Visible

In this activity, you will look at a common reaction that results in a visible change. Please follow along with the experiment and predict the outcomes based on your previous scientific knowledge.



Safety Precautions



- Clean up the work area and wash your hands thoroughly at the end of the activity.

What You Need

50 mL baking soda	250 mL Erlenmeyer flask
75 mL vinegar	candle
100 mL graduated cylinder	matches
	paper towel

SKILL CHECK

Initiating and Planning

Performing and Recording

Analyzing and Interpreting

Communication and Teamwork

What To Do

1. Use the graduated cylinder to measure out 50 mL of baking soda. Place it into the Erlenmeyer flask.
2. Slowly add 75 mL of vinegar to the baking soda in the flask. Cover the jar with paper. Wait about 30 s while the reaction proceeds. In this reaction a gas will be produced, as shown in the figure.
3. Light the candle.
4. Remove the paper from the top of the jar. Carefully pour the gas produced in the flask over the candle flame. Take care not to pour out any of the liquid in the flask.

What Did You Find Out?

1. When have you mixed baking soda and vinegar before? What gas is produced?
2. What happened to the candle flame?
3. Why did this occur?

Key Terms

reactants
products
combustion
neutralization

Check Your Understanding

1. Give three examples of combustion reactions that you encounter in a day. List as many of the reactants and products as you can.
2. Explain one neutralization reaction that occurs in the kitchen.
3. Fishy odours on your hands are caused by a base and can be eliminated with lemon juice. Is this an example of a combustion or a neutralization reaction? Why? What are the reactants?

2.3 Evidence That Change Has Occurred

Chemicals in our environment change constantly. Sometimes we cause these changes. At other times, the changes occur spontaneously without human interference.

During Conduct an Investigation 2–B, you will study many chemical reactions. As you do, look for the evidence that a chemical change has occurred.

Read the contents of the Off the Wall below to learn about various examples of chemical change.



Figure 2.7 Is it magic? No! A chemical change has occurred!



Figure 2.8 Bacteria produce methane gas in wet locations where oxygen is scarce. Methane

gas is often referred to as “marsh gas” because it is present in marshes or bogs. Organic matter, a reactant, undergoes a chemical change that produces methane gas, a product. This chemical reaction also occurs in the intestines of cattle and termites.

Methane gas is a greenhouse gas. In other words, it contributes to the greenhouse effect.



Water Treatment Plant Operator

Water treatment plant operators monitor and operate the equipment used to purify and disinfect water for human consumption. They may work indoors or outdoors, and are usually employed at water, waste water, or sewage treatment plants in small towns or villages. Depending on the size and location of the treatment plant, some may deal directly with the public to handle water quality complaints and emergencies.

The Northern Alberta Institute of Technology (NAIT) offers a one-year Water and Wastewater Technician certificate program. The educational requirement for the program is a high school diploma. NAIT prefers students who have courses in mathematics, biology, chemistry, and physics, but these are not mandatory.

Help Wanted

Water Treatment Plant Operator

Job includes:

- taking water samples
- performing routine laboratory tests for chemical content
- adjusting chemical feed rates of disinfectants
- monitoring flow of water through treatment plant
- operating computerized control systems
- keeping daily records of data readings

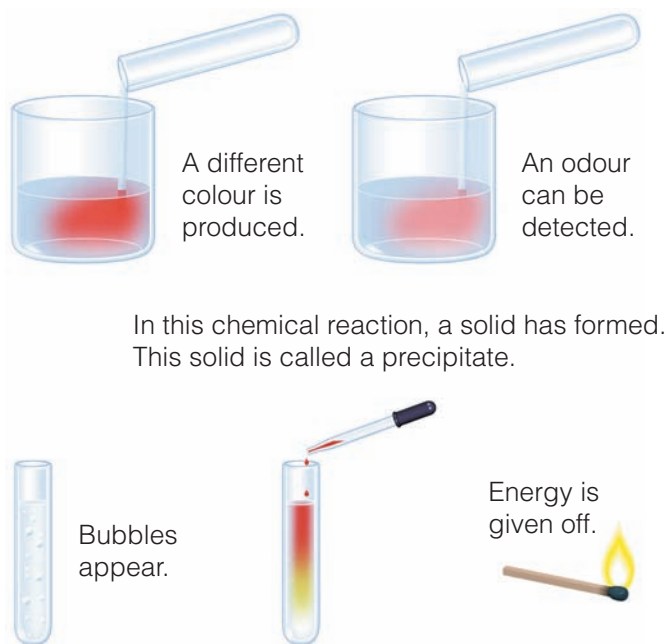


Figure 2.10 Review the five pieces of evidence that indicate a chemical change has occurred.

Evidence of Chemical Change

How can you tell when a chemical change has occurred? Look for some of the following evidence:

- A different colour is produced.
- An odour can be detected.
- Bubbles appear or a gas forms.
- A solid that forms during a chemical reaction is called a **precipitate**.
- Energy is given off in some form — such as light, heat, or sound — or absorbed.

Figure 2.10 shows one example of each.

READING Check ✓

Develop a list of questions you might ask yourself to determine if a chemical change has occurred.

Hot packs are very popular in our cold winters to warm up cold hands and feet. Cold packs are equally useful to relieve a sports injury such as a twisted ankle or bruised knee. Hot packs and cold packs both rely on chemical reactions that involve energy changes in the form of heat.



2.4 Energy and Chemical Reactions

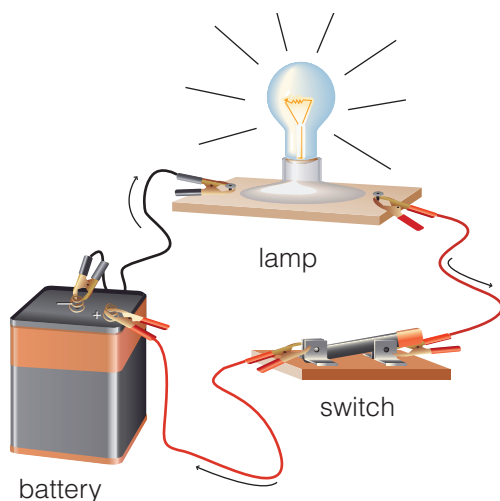


Figure 2.11 Batteries store chemical energy. When a battery is used, a chemical reaction releases electric energy. In this drawing, the electric energy flows through the light bulb, which converts the electric energy to light and heat.

chemical energy $\rightarrow$ electric energy $\rightarrow$ light energy and thermal energy



Figure 2.12 Cold packs are often used by athletes to treat injuries. In cold packs, water and ammonium nitrate are held in separate chambers until the pack is needed. When the chambers are broken, the ammonium nitrate dissolves in the water and absorbs heat. This can make the pack as cold as 0°C . This is an example of an endothermic reaction.

Chemical changes always involve energy changes. Some energy changes are obvious and some are unnoticeable. Fireworks are an example of an obvious energy change. When fireworks explode, a huge amount of heat and light are given off.

Rusting is an example of a chemical reaction involving an unnoticeable energy change. Would you believe that heat is released when metal rusts? The amount is so small that it takes very sensitive equipment to detect it.

Knowledge of energy and energy changes is important to society and industry. For example, gasoline is burned in an automobile engine. This chemical reaction results in forward motion and also produces heat. We need to know how much heat is produced so equipment such as a radiator can be designed to cool the engine. This way, the heat produced by the chemical reaction does not damage the engine.

Gasoline working an engine illustrates an **exothermic reaction** because it **releases energy** into its surroundings. The breakdown (also called the metabolism) of fats and carbohydrates in the human body also releases heat and is exothermic.

Some chemical reactions, like photosynthesis, need to absorb energy in the form of heat or light. Plants need energy from the Sun to produce glucose and oxygen from carbon dioxide and water. **Chemical reactions that absorb energy are called endothermic reactions.**

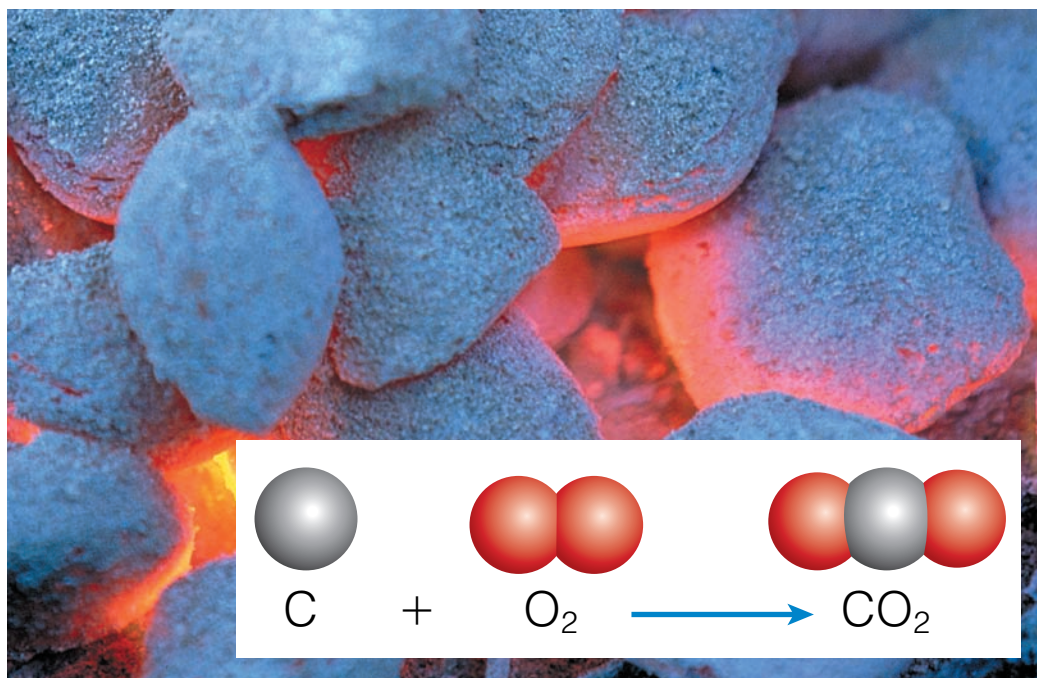
READING
check ✓

Explain the difference between exothermic and endothermic reactions.

Types of Chemical

Getting Ready...

- Name one chemical reaction you are familiar with. What happens during that reaction?
- What are some clues that a chemical reaction has occurred?
- Write one chemical formula that you know. Explain what the symbols mean.
- Why do chemists all over the world understand the short forms for various elements?



Chemical reactions can be so spectacular! Most of us enjoy a colourful show of fireworks. And did you ever have sparklers for a special party?

Some chemical reactions give off light in a variety of colours; others produce sounds.

We use chemical reactions to treat and prevent disease, improve our food supply, purify water, and produce a wide variety of consumer goods. The more we learn about such reactions, the easier it is to use them with minimum risk.

It also helps to know how to describe different reactions. All chemical reactions involve changes in elements and compounds. This

chapter will explain how to record what changes occur to elements and compounds during a reaction.

Just as your friends and family know you by a given name — and often by a nickname, too — most chemical compounds have an official name and a common one. For example, do you ever add *sodium hypochlorite* to your laundry? A chemist knows the compound by that name; you probably know it by its common name — bleach.

In this chapter you will investigate many chemical reactions and learn different ways to describe them. You will also learn some of the universal language chemists use to describe what occurs during a chemical reaction.

Reactions



Sparklers use a chemical reaction.

What You Will Learn

In this chapter you will learn:

- how to represent chemical equations using words and chemical formulas
- how to identify and describe different types of reactions
- how to recognize chemical formulas for some commonly occurring compounds
- why scientists use balanced chemical equations
- how the Law of Conservation of Mass applies to chemical reactions

Why It Is Important

- We live in an age of technology. Chemical reactions make modern technologies possible. Learning how to express and interpret the chemical reactions involved in common processes will enable you to make wiser choices about everything from foods and household products to transportation.

Skills You Will Use

In this chapter you will:

- represent simple chemical reactions using word equations
- identify composition, decomposition, combustion, and neutralization reactions
- demonstrate a knowledge of WHMIS standards by applying proper handling techniques
- name common compounds from chemical formulas
- select and use appropriate methods of communication for ideas, plans, and results

3.1 Word Equations

Here is a summary of what you have learned so far.

- Chemical reactions can be simple (like powering a space shuttle) or complex (like taking an antacid).
- Chemical changes can be obvious (like burning a candle) or less noticeable (like the reaction that makes a cake rise).
- They can occur immediately (like explosives) or more slowly (like the ripening of a banana).

To describe these many different chemical reactions, chemists often use a convenient method known as a **word equation**. A word equation uses words to indicate what changed during the chemical reaction and what is produced.

Word equations are a simple way for chemists to describe chemical changes. They also help us recognize patterns that many reactions have in common.

READING check

Why do chemists use word equations to represent chemical reactions?

Writing Word Equations

Word equations are always written in the same format.

- The left side of the equation lists all the reactants.
- The right side of the equation lists all the products.
- An arrow points from the reactants to the products. It shows that something is produced during the reaction.

all reactants → all products

All the reactants and all the products in a word equation are separated by a plus sign (+).

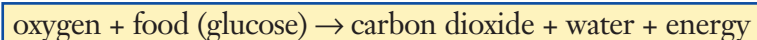
reactant 1 + reactant 2 → product 1 + product 2



Figure 3.1 Limestone caves are formed when the calcium carbonate in the stone combines with acids in rainwater. The product, calcium hydrogen carbonate, is washed away in water. This leaves behind empty caves. Read the next page to find the word equation for this.

Word Equations for Some Chemical Reactions

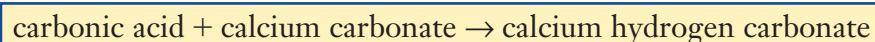
Let's look at some commonly occurring chemical reactions. The first reaction takes place inside your body. It is the key to converting the food you eat into the energy your body can use. The word equation for this reaction is written as:



The equation is read as:

Oxygen and food (glucose) produce carbon dioxide and water and energy.

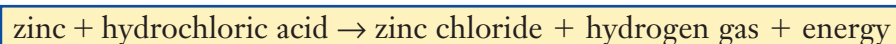
The second word equation describes a common chemical reaction that produces limestone caves:



This equation is read as:

Carbonic acid and calcium carbonate produce calcium hydrogen carbonate.

In Conduct an Investigation 2-B (on page 32), you observed many different chemical reactions. These chemical reactions can all be described using word equations. For example, when you added hydrochloric acid to mossy zinc, you may have noticed that the solution became warmer as the zinc dissolved, and a gas was released. The equation for this reaction is written as:



This equation is read as:

Zinc and hydrochloric acid produce zinc chloride and hydrogen gas and energy.



oxygen + glucose (food) → carbon dioxide + water + energy

Figure 3.2 During cellular respiration, glucose or food is broken down into water, carbon dioxide, and energy in a form that cells can use.

READING check

Do you always need more than one reactant or more than one product to have a chemical reaction?

Check Your Understanding

1. Copy each of the following word equations into your notebook. Beside each, write what you would say as you read the equation.
 - (a) solid zinc + hydrochloric acid → hydrogen gas + zinc chloride
 - (b) ammonium nitrate + sodium hydroxide → ammonium hydroxide + sodium nitrate
 - (c) hydrochloric acid + sodium acetate → acetic acid + sodium chloride

3.2 Types of Reactions

Imagine you are preparing a cake from your favourite recipe, and you discover that you have run out of a few ingredients. If you know there are patterns in the way some of the ingredients go together, you can make substitutions for the missing ingredients. For example, yogurt could be used instead of an egg to increase the lightness of a cake.

READING check

Give two reasons why it is useful for chemists to understand reaction types.

In a similar way, chemists know there are patterns in the way certain elements and compounds go together. They use these patterns to group different types of chemical reactions. Grouping chemical reactions is useful for several reasons. First, it allows chemists to predict the product of an unknown chemical reaction. Second, it helps them to understand the behaviour of substances in a chemical reaction. This section will deal with the four most common reaction types.

Simple Composition Reactions

Figure 3.3 shows what happens when magnesium is ignited in the presence of oxygen. During this reaction, there are two reactants. The first one is the element magnesium. The other is the element oxygen.

If you held the magnesium in the flame, you might observe a reaction taking place. The resulting product would be the compound magnesium oxide. The reaction would also produce energy in the form of bright light. The word equation for this reaction is:

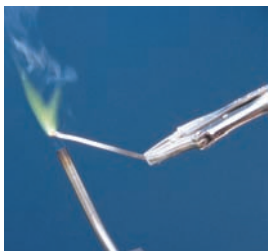
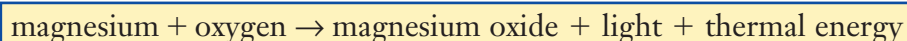
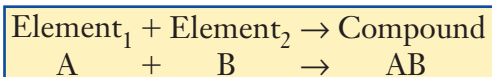


Figure 3.3 When magnesium burns, there is an obvious chemical reaction between two elements. This is an example of a simple composition reaction. What are the reactants? What are the products?

In a **simple composition** reaction, two or more simple elements combine to form a compound. Whenever you see two elements as reactants, you know that the reaction is a simple composition reaction.

In a simple composition reaction, the product is always a compound. If we use the letters A and B to represent two different elements, a simple composition reaction is written in this way:



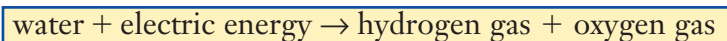
Most simple composition reactions are exothermic, meaning that some form of energy is released. It is wise to know about this type of chemical reaction before trying it in a laboratory.

READING check

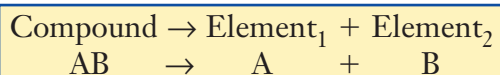
In your own words, define a simple composition reaction.

Simple Decomposition Reactions

During electrolysis, water is broken down into hydrogen gas and oxygen gas. Electric energy is needed to drive this reaction. This makes it an endothermic reaction. The word equation for this reaction is:



A reaction that breaks a compound into its component elements is known as a **simple decomposition** reaction. Whenever you see a compound as the only reactant, you know the reaction is a simple decomposition reaction. In a simple decomposition reaction, the products are always elements. If we use the letters C and D to represent two different elements, a simple decomposition reaction is written in this way:



Most decomposition reactions are endothermic, meaning that energy is absorbed.

You have learned about composition reactions, in which elements are “put together.” You also know something about decomposition reactions, in which compounds are “broken apart.” In Conduct an Investigation 3–A, you will investigate what happens to elements and compounds in these two reaction types.

DidYouKnow?

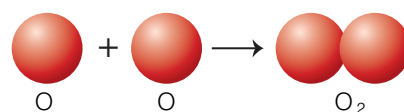
Knowing about reaction types makes it possible for scientists and technologists to predict the products of chemical reactions. Sophisticated lab equipment also helps to protect the investigating scientist.

In the past, experimenters sometimes endangered their health by performing chemical reactions without any advance knowledge of the chemicals that would be produced. English chemist Sir Humphry Davy (1778–1829) frequently inhaled gaseous chemicals he produced in the lab. He experimented with breathing pure carbon dioxide gas, and nearly suffocated himself when inhaling pure hydrogen gas. He also inhaled nitrous oxide gas, which has anaesthetic properties.

Figure 3.6 Sir Humphry Davy



simple composition reaction



simple decomposition reaction

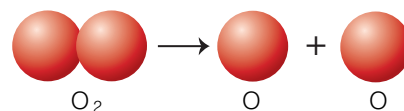


Figure 3.4 A simple composition reaction involves the combination of two elements. A simple decomposition reaction involves the breaking apart of a compound.

Figure 3.5 The process of electrolysis is an example of a simple decomposition reaction. Electric energy is required to drive this reaction, which also makes it an endothermic reaction.



In your own words, define a simple decomposition reaction.



Name another common term for a combustion reaction.

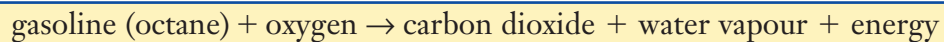
DidYouKnow?

76 L of petroleum can provide the gasoline to drive a vehicle 485 km, or it can produce 24 shirts, 5 m² of carpeting material, 2 automobile tires, 30 m of 1.3 cm diameter rope, 12 windbreakers, 2 tents, 4 sleeping bags, 6 duffel bags, 1 blanket, 4 sweaters, and 15 parkas!

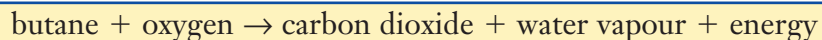
Combustion Reactions

When you light a candle, the candle wax burns in the presence of oxygen to produce carbon dioxide and water. This is one example of a combustion reaction or burning. Combustion always occurs in the presence of oxygen.

Combustion also occurs when you burn fuel to power a car. The word equation for the combustion of gasoline (octane) is:



The same type of reaction occurs when you light a butane lighter. The word equation in this case is:



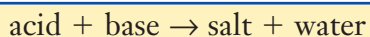
Oil and natural gas are fossil fuels. This means that they come from the accumulated remains of plants and animals from prehistoric times. Fossil fuels form very slowly, which is why many scientists believe we will run out of them. We use them up much more quickly than they can form.



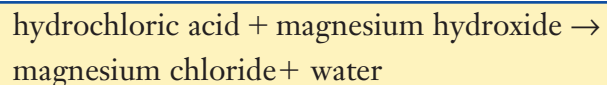
Figure 3.7 Each of these items is a product of the petroleum industry.

Neutralization Reactions

In Chapter 2, you looked at some everyday chemical reactions, such as taking an antacid to relieve acid indigestion. This is an excellent example of a neutralization reaction. In a neutralization reaction, an acid reacts with a base to produce a form of salt and water. The word equation for a neutralization reaction is:



When you take an antacid, the magnesium hydroxide in the tablet reacts with the hydrochloric acid in your stomach. This neutralization reaction is written as:

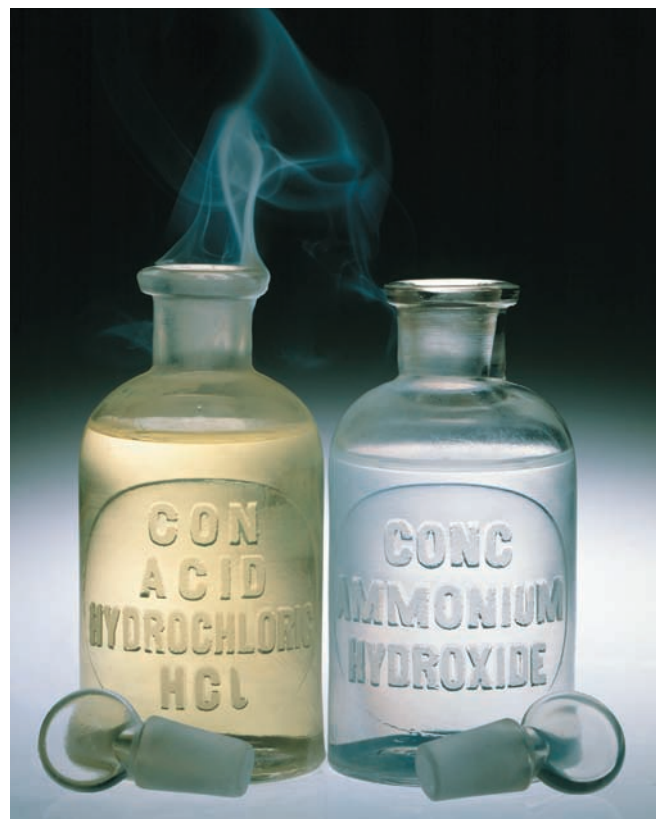


You may be surprised to know how many neutralization reactions occur all around you every day. They are involved in many natural, household, and industrial products and processes.

Figure 3.8 Many acid–base reactions produce water and a form of salt. The white cloud of ammonium chloride seen here is a product of the reaction of hydrochloric acid and ammonia.



In a neutralization reaction, an acid and a base react to form two products. Name the products.



Check Your Understanding

- Read this list of chemical reactions. Identify the ones that show simple composition reactions.
 - methane gas + oxygen gas $\rightarrow$ carbon dioxide + water
 - solid calcium + oxygen gas $\rightarrow$ solid calcium oxide
 - solid sodium chloride $\rightarrow$ solid sodium + chlorine gas
 - solid sodium hydroxide + liquid acetic acid $\rightarrow$ sodium acetate (a salt) + water
 - solid magnesium + oxygen gas $\rightarrow$ magnesium oxide
 - hydrogen gas + oxygen gas $\rightarrow$ water
- Explain the following two equations.
$$A + B \rightarrow AB$$
$$CD \rightarrow C + D$$
 - Which one shows the general format of a simple composition reaction?
 - Which one shows a simple decomposition reaction?

Key Terms

simple composition
simple decomposition

3.3 Formulas for Common Compounds

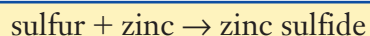
READING check

Why do we need a common international language when we talk about chemical reactions?

What if you spoke a language that no one else could understand? How would you communicate with your friends? Scientists around the world faced this problem when they started sharing information about chemicals and chemical reactions. Because scientists speak many different languages, they created a common language that would allow them to communicate with one another. This common language became the basis of what you are learning about in this chapter.

Chemical Names and Formulas

During a composition reaction you might add zinc powder to sulfur powder and then apply heat. This composition reaction would produce a compound called zinc sulfide. The word equation for this reaction is:



When you write a word equation like this, you are using what is known as the **chemical name** for each of the reactants and products. When you look at the periodic table, you will see the chemical names of all the elements along with their chemical symbols. Each element has a symbol that consists of one capital letter or one capital and one lower-case letter. In this example, you have sulfur (S) and zinc (Zn).

DidYouKnow?

Substances can be in the form of a solid, liquid, or gas. These are known as physical states or states of matter. Many periodic tables show a symbol called a subscript beside each element. This subscript indicates the state in which an element is normally found. For example, the element zinc (Zn) would appear on the periodic table with an _(s), indicating that zinc is normally found as a solid.

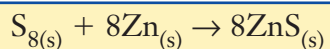
- _(s) to indicate a solid
- _(l) to indicate a liquid
- _(g) to indicate a gas

A fourth subscript — _(aq) — is used to indicate an aqueous solution, which is a substance dissolved in water.

Most chemical compounds are named according to the major elements they contain. Information about the chemical composition and structure of a compound is given by its **chemical formula**. In the above example, the product is the compound zinc sulfide, which has the chemical formula ZnS.

The rule for naming compounds like this is to name the metal first — zinc — followed by the name of the non-metal. The name of the non-metal is changed to “ide” — sulfide.

You could now replace the names of the reactants with their chemical symbols, and the product with its chemical formula. **When writing formulas, remember to use a subscript to indicate the state of the material. In this example, we are dealing with solids, which are shown as _(s).** The chemical equation would look like this:



Common Names

Not everyone is a scientist, so many chemical compounds are often identified by a **common name**. These names are easier to use, and they have existed since before the international naming system was created. You already know many chemicals by their common names. As a student of chemistry, you will learn some chemical names.

Table 3.1 Common and chemical names you already know.

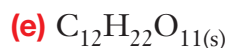
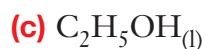
Common Name	Chemical Name	Chemical Formula
alcohol	ethanol	$\text{C}_2\text{H}_5\text{OH}_{(\text{l})}$
antifreeze	ethylene glycol	$\text{CH}_2\text{H}_6\text{O}_{2(\text{l})}$
baking soda	sodium bicarbonate	$\text{NaHCO}_{3(\text{s})}$
chalk, limestone, marble	calcium carbonate	$\text{CaCO}_{3(\text{s})}$
club soda	carbonic acid	$\text{H}_2\text{CO}_{3(\text{l})}$
Epsom salt	magnesium sulfate	$\text{MgSO}_{4(\text{s})}$
laughing gas	dinitrogen monoxide	$\text{N}_2\text{O}_{(\text{g})}$
lye or caustic soda	sodium hydroxide	$\text{NaOH}_{(\text{s})}$
natural gas	methane	$\text{CH}_{4(\text{g})}$
quartz	silicon dioxide	$\text{SiO}_{2(\text{s})}$
rubbing alcohol	isopropyl alcohol	$\text{C}_3\text{H}_7\text{OH}_{(\text{l})}$
table salt	sodium chloride	$\text{NaCl}_{(\text{s})}$
salt substitute	potassium chloride	$\text{KCl}_{(\text{s})}$
sugar	glucose	$\text{C}_6\text{H}_{12}\text{O}_{6(\text{s})}$
	sucrose (table sugar)	$\text{C}_{12}\text{H}_{22}\text{O}_{11(\text{s})}$
water	dihydrogen oxide	$\text{H}_2\text{O}_{(\text{l})}$

Key Terms

chemical name
chemical formula
common name

Check Your Understanding

- Investigate one of the chemicals below. Use your periodic table, Table 3.1, and other chemistry textbooks to describe the chemical as thoroughly as possible.
 - Where might it be used?
 - How many different elements does it contain? How many atoms of each? How many atoms in total?
 - Should it be labelled with a WHMIS symbol? If so, which one(s)?
 - What is its common name? Its chemical name?



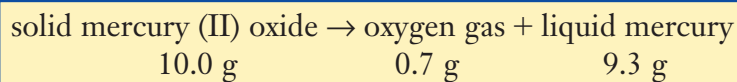
3.4 Chemical Equations

Think back to some of the chemical reactions you have seen in this chapter and the previous one. During the activities, you observed colours changing, bubbles appearing, and precipitates forming. You may also have detected odours being given off and energy being released in the form of light or heat.

In some chemical reactions, matter may appear to be destroyed. Is it really? What happens to the masses of the reactants and the products during a chemical reaction? That is what a French chemist named **Antoine Lavoisier** asked more than 200 years ago.

The Law of Conservation of Mass

In the 1700s, Antoine Lavoisier (1743–1794) placed a carefully measured mass of solid mercury (II) oxide into a sealed flask. When heated, the compound produced liquid mercury and oxygen gas. When this chemical reaction was finished, Lavoisier measured the mass of each of the products. He found that the total mass of the two products was identical to the mass of the reactants. The following word equation shows this reaction. The masses of the reactant and products are shown below the word equation:



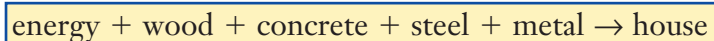
Lavoisier concluded that the atoms in a chemical reaction were just rearranged, not destroyed. This conclusion is known as the **Law of Conservation of Mass**, which states:

In a chemical reaction, matter is not created or destroyed. If you start a reaction with 10 g of reactants, you will end up with 10 g of products.

Figure 3.9 Antoine Lavoisier used precise balances for his investigations. This led to the Law of Conservation of Mass.

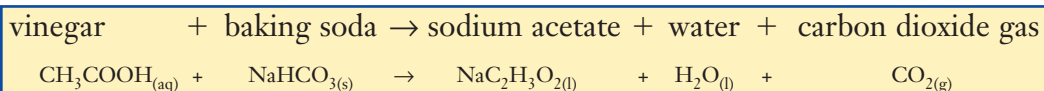
Staying Balanced

When chemists write chemical equations, they do not usually use word equations. When dealing with chemicals, this method is not exact enough. Using a word equation would be like writing a word equation to explain how to build a house.



The builder of such a house would soon have many questions. How much of each material would she need?

Chemists feel the same way. They need to know the exact quantities of chemicals to use when chemical reactions are used in technological processes. Word equations do not state the quantities of the substances needed. They also do not reveal the types of atoms involved. Compare these two equations:



When the chemical formulas of the reactants and products are used, then you can determine exactly what atoms make up the reactants and products. The Math Connect on this page and the one on the next page will help you learn to analyze such equations.

READING check

What is the advantage of using equations with formulas instead of simple word equations?

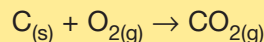
Math CONNECT

Checking the Balance of Simple Equations

A **balanced chemical equation** has the same elements and the same number of atoms of each element on both sides.

Sample Problem

The equation that follows shows what happens when carbon and oxygen gas go through a chemical reaction. Is the equation balanced?



Steps in Solving It

1. *What elements are on the reactant side?*
The letters shown are C and O. If you cannot remember what these stand for, check a copy of the periodic table. C is the element carbon. O is the element oxygen.
2. *How many atoms of each element are on the reactant side?*
There is no subscript beside carbon. This means that there is only one atom of carbon.

The subscript 2 (₂) beside oxygen tells you there are 2 atoms of oxygen.

3. *What elements are on the product side?*
Again, there is a C and an O.
4. *How many atoms of each element are on the product side?*
There is no subscript with C, so there is only one atom of carbon. O has a subscript — 2. So there are two atoms of oxygen.
5. *Is this equation balanced?*
reactant side = 1 atom of carbon, 2 of oxygen
product side = 1 atom of carbon, 2 of oxygen
Yes, the equation is balanced.

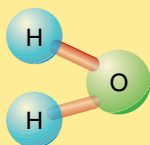
Solve These

Use the example above to help you check whether the following equations are balanced. Be sure to show your thinking and calculations.

1. $\text{Cl}_{2(\text{g})} + \text{H}_{2(\text{g})} \rightarrow 2\text{HCl}_{(\text{g})}$
2. $\text{FeO}_{(\text{s})} \rightarrow \text{Fe}_{(\text{s})} + \text{O}_{2(\text{g})}$

Checking the Balance of Complex Equations

Water electrolysis is an example of a simple decomposition reaction. You observed such a reaction during Conduct an Investigation 2–B on page 32. Although the reaction is simple, the equation that represents it is not so simple. This Math Connect will show you how to read and check the balance of such equations.

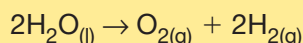


This molecular model of a water molecule ($\text{H}_2\text{O}_{(l)}$) clearly shows there are two hydrogen (H) atoms and one oxygen (O) atom in each molecule of water.

The equation for showing the decomposition of water is $2\text{H}_2\text{O}_{(l)} \rightarrow \text{O}_{2(g)} + 2\text{H}_{2(g)}$. In such equations, the numbers in front of the chemical formulas are called coefficients. **Coefficients** are used to balance those individual atoms. In $2\text{H}_2\text{O}_{(l)} \rightarrow \text{O}_{2(g)} + 2\text{H}_{2(g)}$, the 2 in front of the H is a coefficient. It tells how many H_2 molecules are present.

Sample Problem

Read the equation. Identify and count the atoms on each side. Is it balanced?



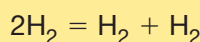
Steps in Solving It

1. What elements are on the reactant side?

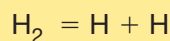
The letters shown are H and O. If you cannot remember what they stand for, check a copy of the periodic table. They are hydrogen and oxygen.

2. How many atoms of each element are on the reactant side?

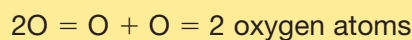
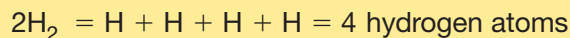
The 2 in front of $\text{H}_2\text{O}_{(l)}$ makes this tricky. The large 2 means that there are 2 H_2 and 2 O. Let's deal with each one separately.



The subscript 2 (₂) beside hydrogen tells that there are two hydrogen atoms in each H_2 molecule.



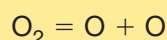
And there are two of these in the chemical name. So



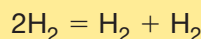
3. What elements are on the product side?

Again, you see H and O. These are hydrogen and oxygen.

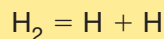
4. How many atoms of each element are on the product side?



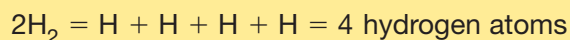
The 2 in front of $\text{H}_{2(g)}$ makes this tricky. The large 2 means that there are 2 $\text{H}_{2(g)}$. Let's deal with each one separately.



The subscript 2 (₂) beside hydrogen tells that there are two hydrogen atoms in each H_2 molecule.



And there are two of these in the chemical name. So



5. Is this equation balanced?

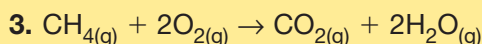
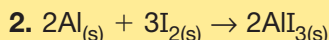
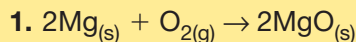
reactant side = 2 oxygen atoms,
4 hydrogen atoms

product side = 2 oxygen atoms,
4 hydrogen atoms

Yes, the equation is balanced.

Solve These

Use the example above to help you check whether the following equations are balanced. Be sure to show your thinking and calculations.



4

Reactions and the

Getting Ready...

- Why are coal, oil, and natural gas called “fossil fuels”?
- Name the type of reaction that gives off heat or light.
- What are some clues that a combustion reaction has occurred?
- What type of chemical reaction occurs when you mix vinegar and baking soda?



What do these pictures have to do with the study of chemistry?

To answer, examine the photograph of the cars and trucks driving over the steel bridge. Chemistry has allowed us to harness the energy stored in oil, natural gas, and other fossil fuels to power our vehicles and industries.

Chemistry also plays an important role in the steel bridge. Chemical reactions are used to make the steel. Chemical reactions that form rust are also at work trying to weaken the structure.

Even something as natural as rain has a chemical connection. In some parts of Canada, pollution from

combustion reactions has created acid deposition. The picture at the top left of page 59 shows a forest damaged in this way.

In this chapter, you will explore chemical reactions between acids and bases and learn how we use this knowledge to combat acid deposition.

You will also investigate how combustion reactions lead to global climate change and acid deposition, and you will demonstrate the environmental conditions that cause the corrosion of steel. In addition, you will identify technologies that protect steel against corrosion.

Environment



What You Will Learn

In this chapter you will learn:

- how to describe changes to reactants and products in fossil fuel combustion reactions
- how the products of fossil fuel combustion affect our environment
- how to explain common acid–base neutralization reactions
- how technologies can reduce the emissions that affect our environment
- how to describe changes to reactants and products in rusting reactions

Why It Is Important

- Some of the products and conveniences you enjoy can negatively affect the environment. What if you had to give up some of the conveniences you have come to depend on? Knowing more about chemical reactions helps you make choices that are good for you — and for the environment.

Skills You Will Use

In this chapter you will:

- plan investigations of questions, ideas, problems, and issues
- carry out procedures using a broad range of tools and techniques
- demonstrate a knowledge of WHMIS standards by handling chemicals and other materials properly
- compile, analyze, and display information in a variety of formats
- work as part of a team to address problems and communicate information



An explosion in Halifax on December 6, 1917, was one of largest, most violent human-made, energy-releasing chemical reactions in history. This chemical reaction killed nearly 2000 people and injured about 9000 others.

4.1 Fossil Fuel Combustion



Figure 4.1 Fossil fuels provide the energy to power cars, and provide heat to appliances such as this gas stove burner.



Name the products of a combustion reaction.

How many of the following things do you do each day?

- switch on a light bulb or use an electric hair dryer
- drive a car or get a ride on a bus
- heat your home with a natural gas furnace
- use natural gas to heat the water for your shower or bath

If you do any or all of these, you are relying on the combustion or burning of fossil fuels.

Chapter 3 discussed that fossil fuels are energy resources from the decayed remains of ancient plants and animals.

In most cases, the energy stored in fossil fuels must be transformed into other forms of energy before it can be used. Combustion, or burning, is the most common way to transform this stored energy.

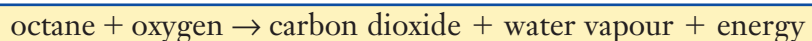
During a combustion reaction, a fossil fuel reacts rapidly with oxygen gas. The products of this combustion reaction are carbon dioxide gas ($\text{CO}_{2(g)}$) and water vapour ($\text{H}_2\text{O}_{(g)}$).

Like all exothermic reactions, the combustion of fossil fuels gives off energy. In a car engine, some of the chemical energy released is transformed into kinetic energy to turn the wheels of the car. If you feel a car hood after a long trip, you will know what most of the energy is given off as heat.

The following chemical equation describes the combustion of octane, a component of gasoline:



The word equation for this reaction is:



In the following activity, you will investigate changes to the reactants and products in a common fossil fuel combustion reaction.

Evidence of a Combustion Reaction

When you light a candle, you observe fossil fuel combustion as the candle wax (made from fossil fuels) burns. In this activity, you will identify more evidence of fossil fuel combustion.

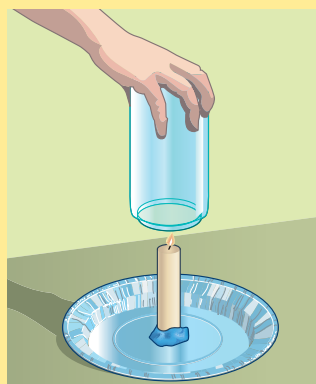
Safety Precautions



- Tie back long hair and secure any loose clothing.
- Clean up the work area and wash your hands thoroughly at the end of the activity.

What You Need

small ball of modelling clay
candle
pie plate with high sides
matches
glass jar
water
food colouring
ruler



What to Do



Part 1

1. Use the modelling clay to anchor the candle in the bottom of the pie plate.

2. Light the candle with a match.
3. Place the glass jar over top of the burning candle.
4. Carefully observe the candle as well as the glass jar. Observe any evidence that a combustion reaction occurred in the glass jar.

Part 2

5. Remove the glass jar from over the candle.
6. Add enough coloured water to cover the bottom of the pie plate to a depth of 1 cm.
7. Light the candle.
8. Place the glass jar over top of the burning candle.
9. Make sure you observe the candle as well as the events that occur in the glass jar. Observe any evidence that a combustion reaction occurred in the glass jar.

Incomplete Combustion

If fossil fuels do not burn completely, they undergo a chemical reaction called incomplete combustion. Incomplete combustion occurs when there is not enough oxygen available. For example, it may occur in a blocked chimney, an inefficient furnace, or a car engine running in a closed garage.

Complete combustion produces three products: carbon dioxide ($\text{CO}_{2(g)}$), water vapour ($\text{H}_2\text{O}_{(g)}$), and energy. Incomplete combustion produces five products: carbon monoxide ($\text{CO}_{(g)}$), carbon ($\text{C}_{(s)}$), carbon dioxide, water vapour, and energy.

Carbon monoxide is an odourless, colourless gas that is extremely poisonous. It combines with blood in the bloodstream to starve the body of oxygen, causing death. The carbon produced in an incomplete combustion reaction causes the black coating (soot) you see in fireplaces.

Greenhouse Gases

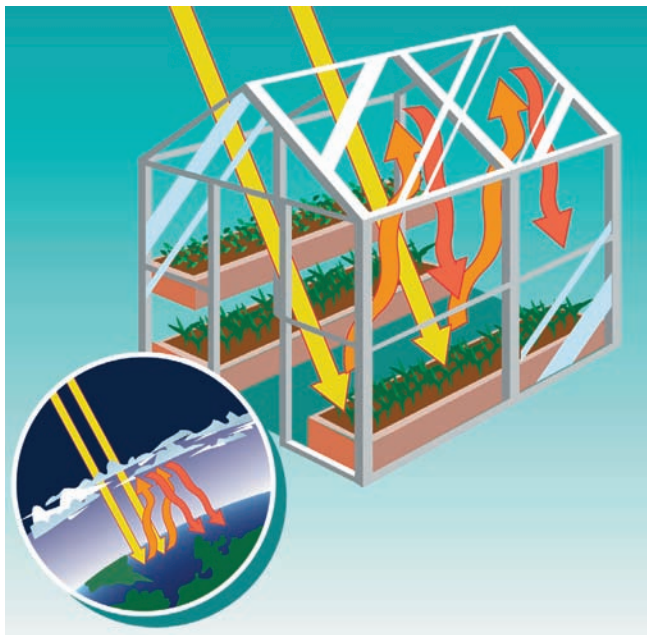


Figure 4.2 A greenhouse prevents heat from escaping into the atmosphere. Greenhouse gases work the same way to trap heat in Earth's atmosphere.

We use fossil fuels to heat homes, provide energy for industry, and generate electric energy. Our cars and trucks burn millions of litres of gasoline daily. All of these combustion reactions release millions of tonnes of carbon dioxide into the atmosphere. What effect does all this extra carbon dioxide have on Earth's atmosphere?

Most scientists agree that carbon dioxide is important to life on Earth. All living creatures, including plants, produce it naturally. In fact, all living things produce carbon dioxide, water vapour, and methane. These are known as natural **greenhouse gases**.

Greenhouse gases trap heat in Earth's atmosphere and prevent heat from escaping into space. They also help to maintain the average global temperature of $+15^{\circ}\text{C}$. Without them, the average global temperature would be much colder — about -18°C !

Global Climate Change

Throughout Earth's history, there have been intervals of warming and cooling. Scientists are concerned about the current long-term trend toward **global warming**. Accompanying this trend is a dramatic increase in extreme climatic events. Glaciers are melting. Sea levels are rising. Climate zones are shifting.

The real threat is the *rate* at which **climate change** is occurring. Climates are changing much faster than in Earth's past. Scientists point out that many species of organisms will not be able to adapt to such fast changes. These organisms may become extinct.

Some people believe that climate change is a natural phenomenon. They note that Earth has gone through many ice ages with shorter periods of warming between the ice ages. The Earth and its species will adjust.

In the following investigation, you will consider both sides of the global warming issue.



In your own words, explain global climate change.

- ☀ Initiating and Planning
- ☀ Performing and Recording
- ☀ Analyzing and Interpreting
- ☀ Communication and Teamwork



Glacier high above Peru in the Andes Mountains, 1978



The same glacier, 2000

Climate Change: What is Your Position?

Think About It

There is abundant evidence that the global climate has warmed significantly during the last century. Scientists point out that this is due to the increase of two greenhouse gases: carbon dioxide and methane.

- The burning of fossil fuels such as coal and gasoline has released tremendous amounts of carbon dioxide.
- Human activities such as burning vegetation, landfiling garbage, growing rice in large paddies, and keeping large amounts of cattle together in feedlots release large quantities of methane.

The level of these and other greenhouse gases released by human activities appears to be causing the current rapid global warming.

The Earth has experienced natural variations in climate throughout its history. In fact, the greenhouse effect maintains the temperature of the Earth at a level suitable for life to exist.

Natural climate changes occur over a relatively long period of time. Because the change is gradual, organisms adapt to the new temperature and precipitation patterns.

Today, many scientists are concerned about the rate of change. Global temperatures are changing much faster than in the past. Will species have enough time to adapt to the new climate?

Some skeptics argue that there is nothing to worry about. Warmer temperatures will make Earth more hospitable for humans and other organisms. They also question the reliability of the climate models that scientists use to make predictions about climate change. What do you think?

What to Do

- 1 Work in teams of three or four students.
- 2 Each member of the group should use an approved search engine to locate:
 - one web site that *supports* the scientific viewpoint that human activities are directly related to climate change; and
 - one web site that *refutes* the scientific viewpoint that human activities are directly related to climate change.
- 3 Evaluate and summarize the information found on each web site.

Analyze

1. As a group, compare the information that you found supporting and refuting the scientific view that human activity is responsible for the rapid increase in global temperatures. Use a chart or other tool to organize the data that you collect.

Air Pollution

Carbon dioxide is not the only chemical substance that human activities add to the air. Many other chemicals in the air cause harm to living things or the environment. These chemicals are called **pollutants**.

READING

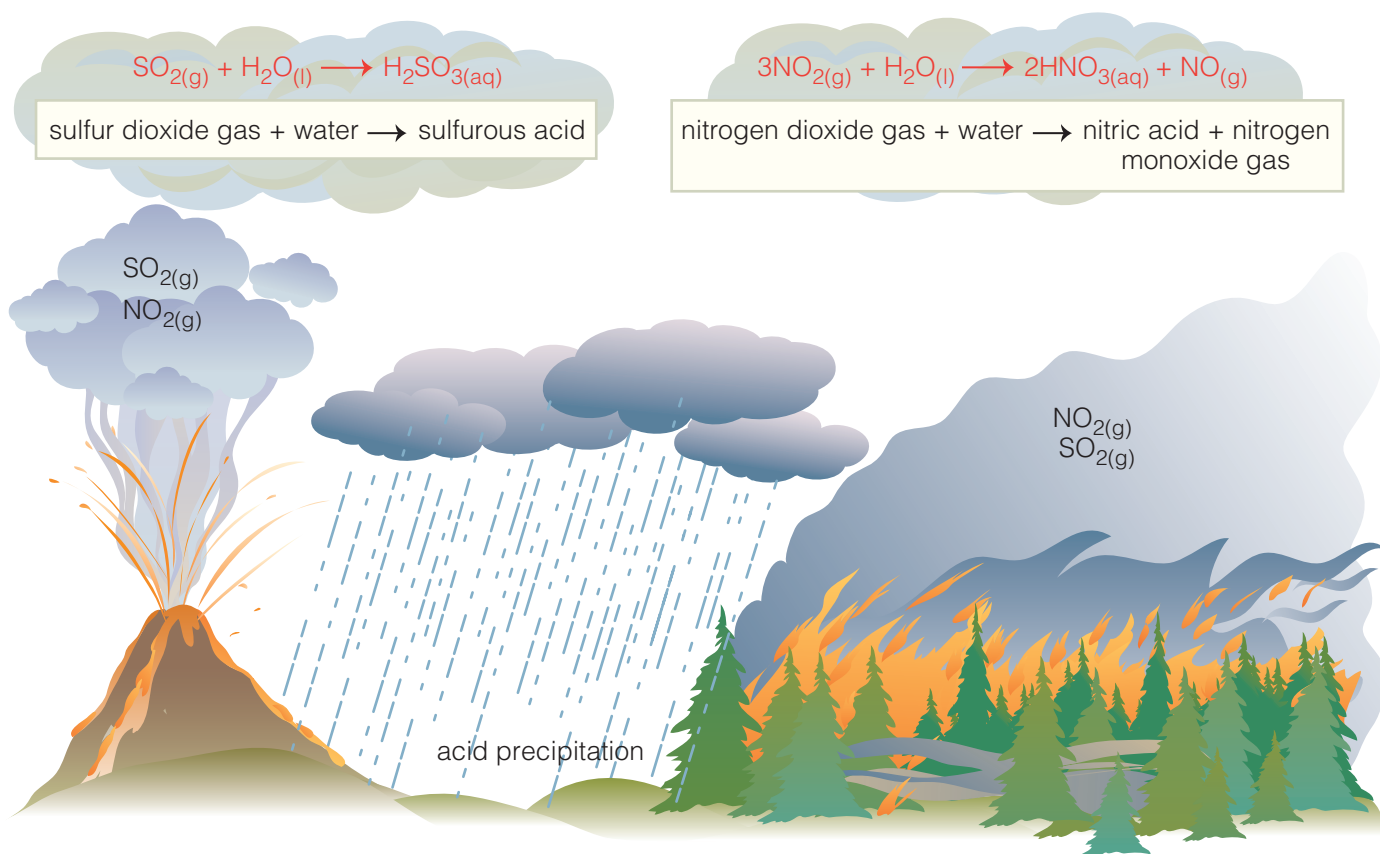
check

In your own words, explain how acid deposition is produced.

Air pollutants can cause problems such as asthma, headaches, and irritation of the eyes, nose, and throat. They can also aggravate heart and lung conditions.

Even worse, air pollutants can combine with each other to produce **acid precipitation** or rainwater that has become more acidic. The more accurate term for this phenomenon is **acid deposition** since dry matter, as well as precipitation, causes acids. Study Figure 4.3 and identify the different sources of pollutants that contribute to acid deposition. Notice the chemical reactions involved.

Figure 4.3A Natural Formation of Acid Deposition.

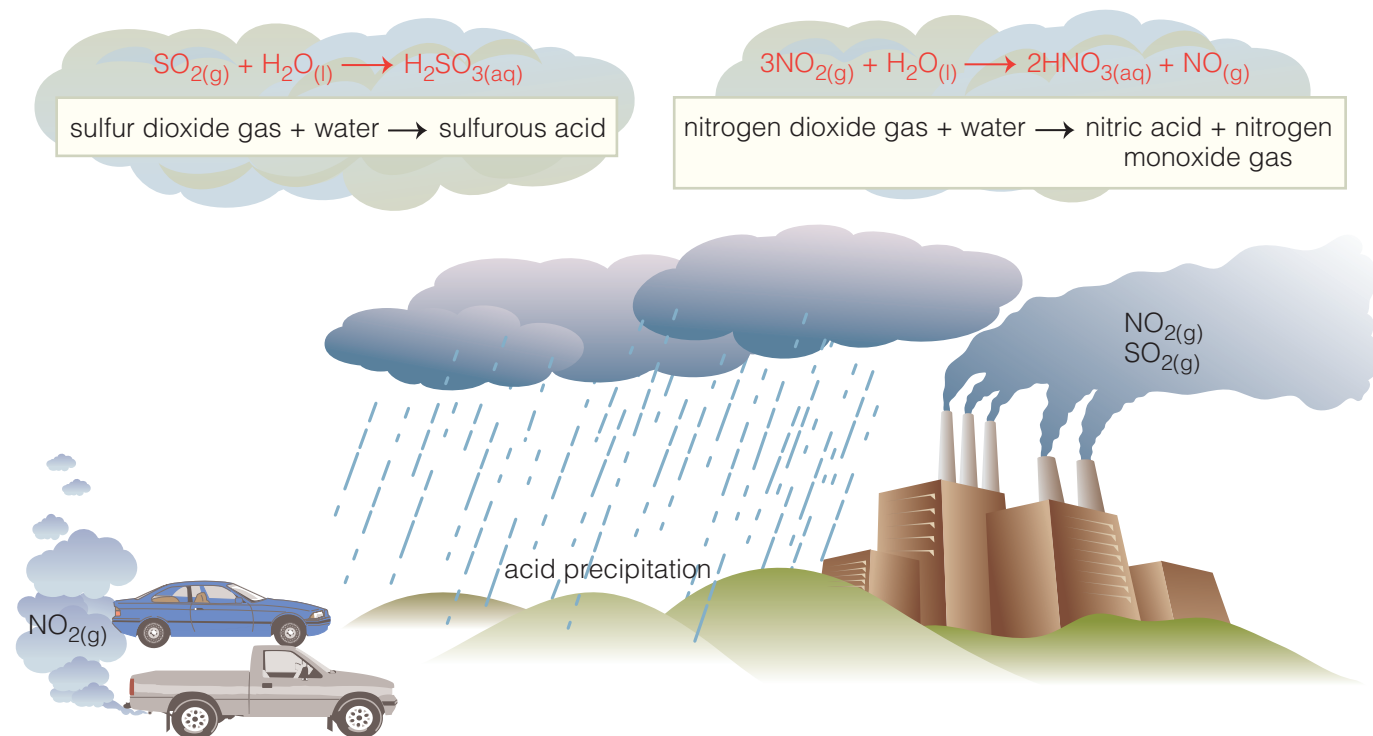


When volcanoes erupt, they release carbon dioxide and water vapour. They also release **sulfur dioxide ($\text{SO}_{2(g)}$)** and **nitrous oxides ($\text{NO}_{x(g)}$)** into the atmosphere. These gases mix with water vapour in the clouds to form diluted sulfurous and nitric acids.

Forest fires release carbon dioxide and water vapour. The burning of wood also releases nitrogen dioxide ($\text{NO}_{2(g)}$) and some sulfur dioxide ($\text{SO}_{2(g)}$) into the atmosphere. Both of these gases react with rainwater to make acid deposition.

Figure 4.3B Human Contribution to Acid Deposition

Like air pollutants, acid deposition comes from both natural sources and human activities. Study the illustrations here and on page 64 to identify both natural and human-made causes of acid deposition.



In a similar way, nitrogen oxides emitted from car exhaust react with rainwater to form nitric acid.

Industrial smokestacks release sulfur dioxide and nitrogen oxides, which react with rainwater to make acid deposition.

Check Your Understanding

1. List three natural greenhouse gases.
2. Describe three negative effects of global climate change.
3. What are the major contributors to acid deposition?
4. Name two gas emissions that are linked to acid deposition.

Key Terms

greenhouse gases
global warming
climate change
pollutants
acid precipitation
acid deposition
sulfur dioxide
($\text{SO}_{2(g)}$)
nitrous oxides
($\text{NO}_{x(g)}$)

4.2 Acid–Base Reactions

READING

Check

Explain a neutralization reaction that occurs in your digestive system.

Chapter 3 discussed what happens when an acid and a base are combined. This produces a neutralization reaction. During this neutralization reaction, the acid and base react together to form two new chemical compounds: a salt and water.

Neutralization reactions do not just happen in labs. They also happen in your body. Study Figure 4.4 — read both the caption and labels — to see how your digestive system uses neutralization reactions.

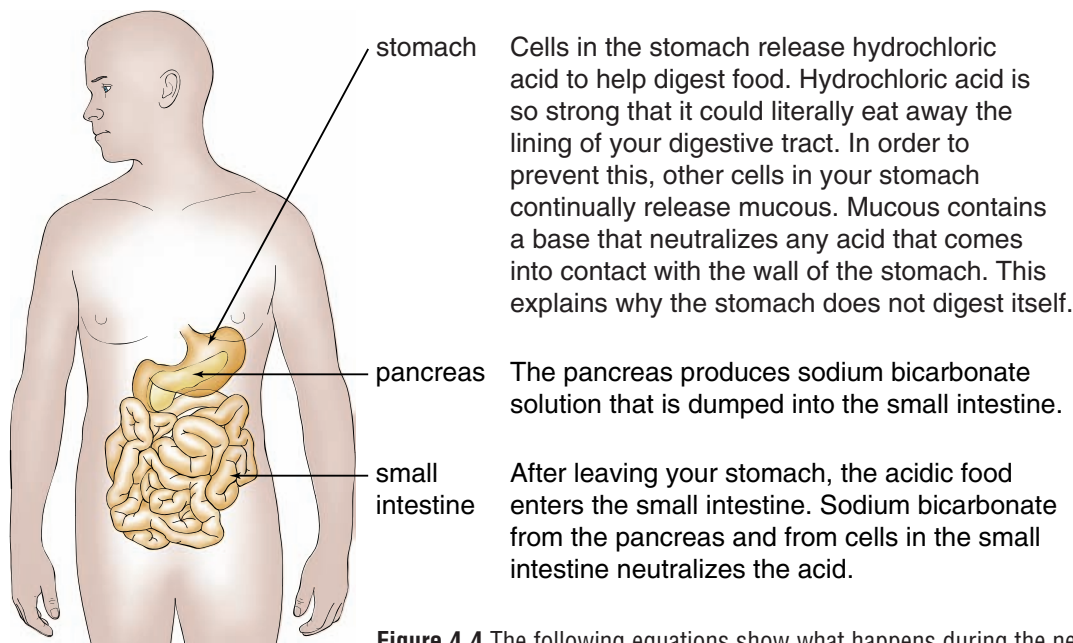


Figure 4.4 The following equations show what happens during the neutralization reaction in your digestive system. What product does this reaction produce?

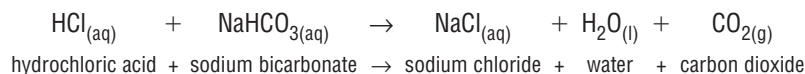


Figure 4.5 Calcium carbonate is released into a lake affected by acid precipitation. This neutralizes the lake water. Such liming is not a permanent solution to the effects of acid precipitation. Why do you think this is so?

Acids and Bases in the Environment

Natural limestone ($\text{CaCO}_{3(\text{s})}$) is a base used by industry. This base is also used to neutralize lakes that have been acidified by acid precipitation. This technique is called **liming**.

During liming, calcium carbonate ($\text{CaCO}_{3(\text{s})}$) is added to the waters of an acid lake. The calcium carbonate neutralizes the acidic water, thus raising the pH of the lake water.

READING

Check

Explain the benefit of adding lime to lakes.

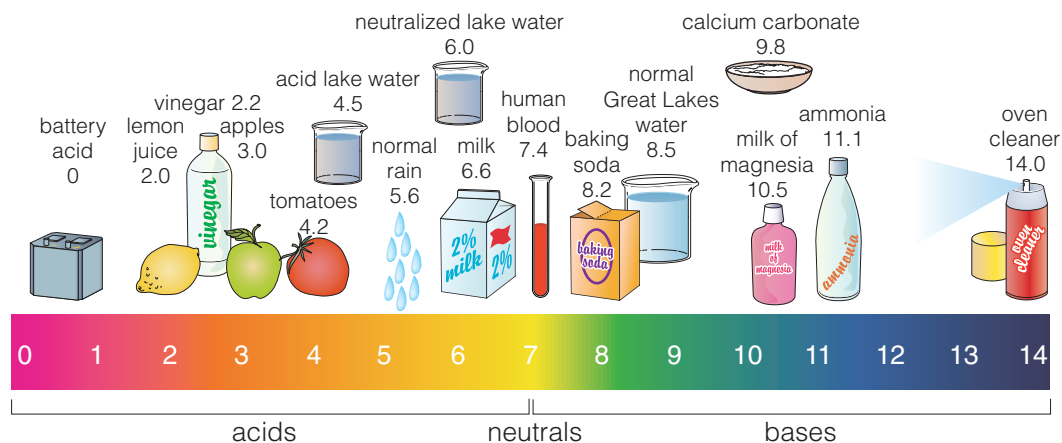


Figure 4.6 The pH scale tells you how acidic or basic a substance is. Identify the pH levels of an acid lake, calcium carbonate, and a neutralized lake.

Reducing Emissions

A better way to reduce acid deposition is to prevent it from forming. In Figure 4.3 on page 64–65, you learned that sulfur and nitrogen oxides are responsible for the high levels of acid deposition. Sulfur dioxide is a major by-product of industrial processes. It is also a product in the combustion of fossil fuels like natural gas.

One way to reduce sulfur emissions is to remove the sulfur from the fuel before the fuel is burned. Figure 4.7 shows one place in Alberta where this is done. Turn the page to learn *how* it is done.

DidYouKnow?

The coal mined in Alberta is valued for its low sulfur content.

Figure 4.7 At gas plants such as this one in Waterton, Alberta, hydrogen sulfide has been removed from natural gas. The by-product of this chemical reaction is sulfur.



READING Check

Name two sources of sulfur dioxide emissions.

READING Check

Explain why sulfur is removed from natural gas.

Removing Sulfur from Fuel

Most scientists agree that using sweet natural gas is one way to reduce emissions that produce acid precipitation. You may have noticed large yellow piles of sulfur at some natural gas plants in Alberta. This sulfur has been removed from sour natural gas before it is sent to consumers. Since sulfur emissions are one of the main causes of acid precipitation, removing sulfur from the fuel *before* burning it will reduce acid precipitation.

Sulfur Scrubbing

The coal produced in some areas of the country contains sulfur. **When this coal is burned, the sulfurous emissions can return to Earth as acid precipitation. To help prevent this, a device known as a scrubber is installed on the smokestacks of many industrial plants that burn coal.**

Scrubbing is a chemical and physical process that removes sulfur compounds formed during the combustion of coal. Look at Figure 4.8 to see how this is done. Scrubbers remove the sulfur from the smokestack before it reaches the atmosphere. Study the equations in the figure and trace what happens to the coal, limestone, and air that enter the furnace.

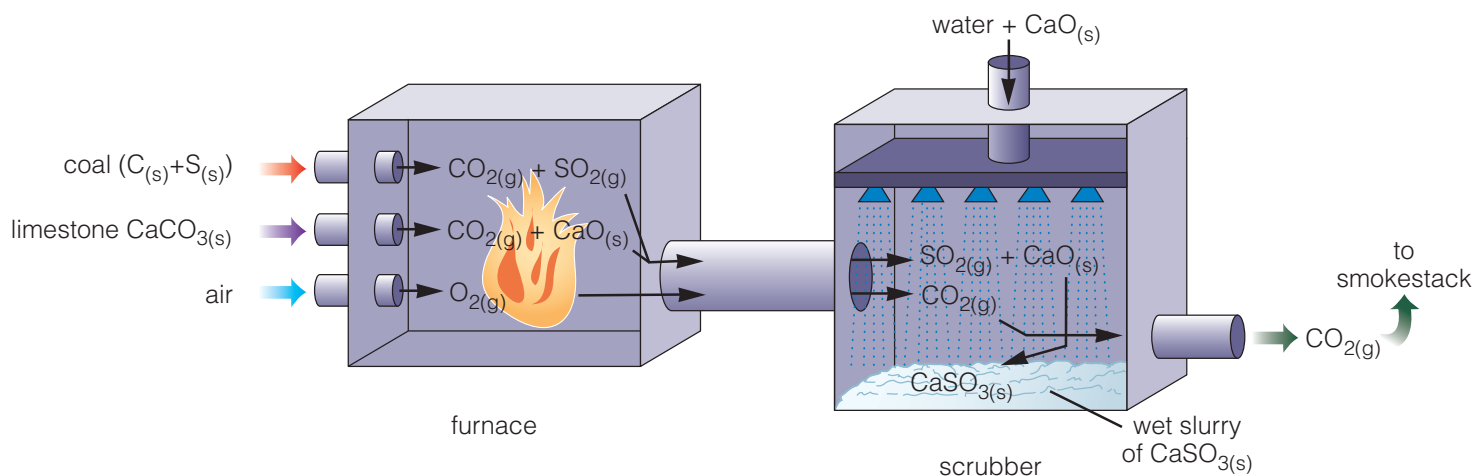


Figure 4.8 In the final stages of scrubbing, calcium oxide ($\text{CaO}_{(s)}$) is injected. It reacts with sulfur dioxide gas ($\text{SO}_{2(g)}$) to form a wet slurry of calcium sulfite ($\text{CaSO}_{3(s)}$). Calcium sulfite forms the sludge that is washed away with water.

Key Terms

liming
scrubber

Check Your Understanding

1. Write out the general chemical equation for a neutralization chemical reaction.
2. Describe how driving a car contributes to the problem of acid precipitation.
3. Explain how liming can raise the pH of a lake affected by acid precipitation.
4. Why is it important to remove sulfur from coal and natural gas before they are burned?

4.3 Corrosion of Iron



Figure 4.9 Iron and steel will rust when exposed to oxygen and moisture.

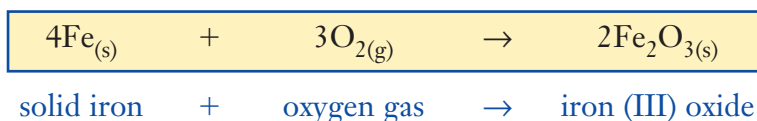


Figure 4.10 Rusting is an example of a simple composition chemical reaction. Check the information on page 44 in Chapter 3 to review this reaction type.

Corrosion refers to any process that chemically breaks down or degrades metal. The best-known case of corrosion is the **rusting** of iron. Most metals will rust, but the corrosion of iron and steel is a serious problem because these metals are widely used. **As rust flakes off an iron structure, more metal is exposed to the environment. The structure is weakened over time, as more and more metal is exposed.**

Have you ever left a metal tool or toy outside and found that it rusted and broke? **If so, you might think rust is a decomposition reaction. In fact, rusting is a simple composition reaction.**

Examine the chemical reaction for the corrosion of iron, shown below. Iron is reacting with oxygen to produce iron oxide, which is commonly called rust. Although it is not included in this chemical equation, **water is also required for rusting to occur.**



Scientists and engineers are concerned about the corrosion of steel. Unless rusting is prevented, it will weaken an iron or steel structure. The rusting of the steel used to reinforce concrete in bridges and buildings can cause these structures to fail and collapse. Repairing or replacing these structures costs millions of dollars.

READING check

Explain how rusting is a simple composition reaction. What are the reactants? What is the product?

DidYouKnow?

Gold is one metal that does not corrode. This is one reason why many gold objects have survived for thousands of years.

4.4 Solving the Problem of Corrosion



Figure 4.11 When oxygen and water contact the surface, metals such as iron and steel will rust.



Figure 4.12 Paint can protect steel from corrosion. This bridge has been cleaned with high-pressure jets of water to remove loose paint, rust, and other loose debris. The cleaned surface was then given a fresh coat of protective paint.

Have you ever noticed that metal parts of a car that have been damaged in a collision are more susceptible to rusting? Why would some metal parts of the car rust while others remain rust free? The answer is in the way metals are protected.

Painting — One way to protect metals is to give them a protective coating of paint. Painting over the metal prevents oxygen and water from reaching the surface of the metal. This explains why steel beams used in office buildings or on bridges are often painted with red paint.

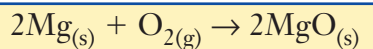
Painting, however, may not be the best way to permanently protect iron or steel. Once the paint chips off, water and oxygen are free to attack the steel, and rust blisters grow rapidly.

Galvanizing — Another way to protect steel or iron is to use a process called **galvanization**. Galvanization is the process of applying a zinc coating to iron or steel. This involves immersing the metal in a bath containing molten zinc. This process coats the metal with a layer of zinc that provides a protective barrier between the metal and the environment. The galvanized coating adheres to the base metal and resists chipping and cracking. In fact, the zinc coating actually becomes part of the surface of the steel.

Some galvanized nails also have a polymer coating to increase the protection against rusting.

Sacrificial Metal — A third method of protecting iron involves the use of a **sacrificial metal**. If you could look closely at an underground oil pipeline, you would notice bars or ingots of a second metal, such as magnesium, buried with the pipeline. The magnesium will corrode but the pipeline will not. In this case the magnesium is sacrificed in order to protect the pipeline.

The equation for this is:



It is much easier and cheaper to replace the bars of magnesium than it is to repair a ruptured pipeline.

READING
check ✓

Describe three methods of protecting metals from corrosion.

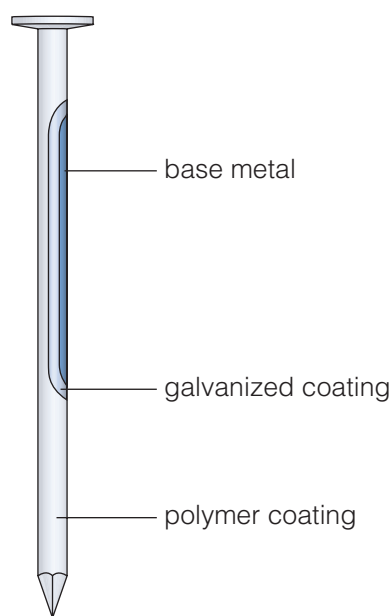


Figure 4.13 Galvanized nails resist rust for many years.

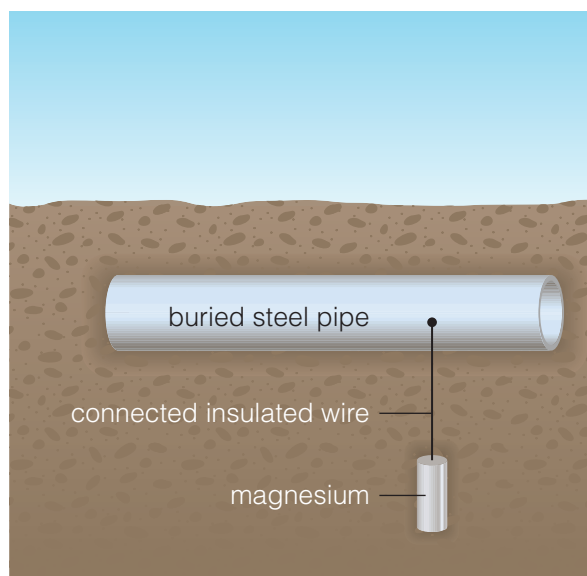


Figure 4.14 Magnesium ($\text{Mg}_{(s)}$) is sacrificed to protect steel pipelines that contain iron ($\text{Fe}_{(s)}$). Magnesium, zinc, and aluminum are commonly used as sacrificial metals.

Check Your Understanding

1. Explain why the chemical reaction for the rusting of steel is similar to the chemical reaction for the combustion of a fossil fuel.
2. How does painting a steel beam protect it from rusting?
3. How does dipping nails into molten zinc help prevent them from corroding?

Key Terms

galvanization
sacrificial metal